

CO-ORDINATE GEOMETRY

Synopsis

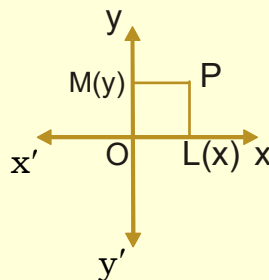
Date: 16-08-2010

Introduction:

COORDINATE PLANE

Let A,B be two points in a plane. Generally the distance between A and B denoted by AB. The line segment joining A and B is denoted by $\overline{AB}$. The ray from A and passing through B denoted by $\overrightarrow{AB}$. The ray from A and passing through B is denoted by $\overleftarrow{AB}$. The line passing through A and B is denoted by $\overleftrightarrow{AB}$.

Let $\overleftrightarrow{x'x}$ and $\overleftrightarrow{y'y}$ be two mutually perpendicular coordinate lines in a plane intersecting at O. The point O is called origin. The line $\overleftrightarrow{x'ox}$ is called x - axis as horizontal line and y- axis as vertical line. Let p be a point in the plane. Let L,M be the projections (feet of the perpendiculars) of P on x - axis and y-axis respectively. Let x,y be the real numbers assigned to L,M on the lines $\overleftrightarrow{x'ox}, \overleftrightarrow{y'oy}$ respectively.



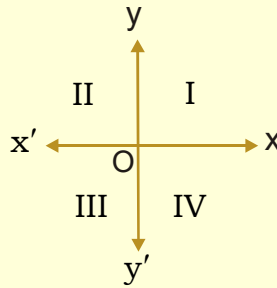
Then x is called x - coordinate of P and y is called y - coordinate of P. The point P is denoted by (x,y).

Every point in the plane can be represented by two numbers x,y (coordinates) and every pair of real numbers determine a point in the plane.

x- axis and y-axis are called coordinate axes and the plane is called coordinate plane. The coordinate axes divide the plane into four equal parts. Each part is called a quadrant. The regions xoy, yo x', x'oy', y'ox are called I,II,III,IV quadrants respectively.

Let P(x,y) be a point in the coordinate plane. Then

- 1) P lies in I quadrant iff $x > 0, y > 0$
- 2) P lies in II quadrant iff $x < 0, y > 0$
- 3) P lies in III quadrant iff $x < 0, y < 0$
- 4) P lies in IV quadrant iff $x > 0, y < 0$
- 5) P lies in the x - axis iff $y = 0$.
- 6) P lies in the y - axis iff $x = 0$.
- 7) If P = O, then $x = 0, y = 0$ i.e., the coordinates of the origin O are (0,0) .



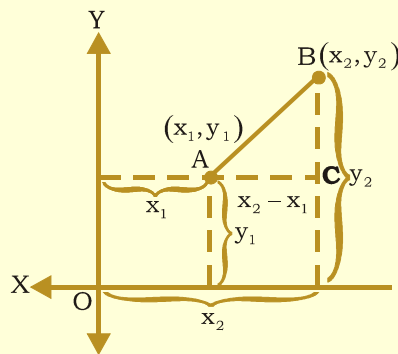
Note :

Let $A(x_1, y_1), B(x_2, y_2)$ be two points in the coordinate plane. Then (i) $\overline{AB}$ is a horizontal line iff $y_1 = y_2$ and (ii) $\overline{AB}$ is a vertical line iff $x_1 = x_2$.

We state some results without proofs in this chapter which were learnt in previous classes.

Distance between two points

Let $A(x_1, y_1), B(x_2, y_2)$ be any two points on a line not parallel to the axes. From the adjacent figure we have the right angle triangle ABC.



$$AB^2 = AC^2 + BC^2$$

$$\text{But } AC = x_2 - x_1, BC = y_2 - y_1$$

$$\therefore AB^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

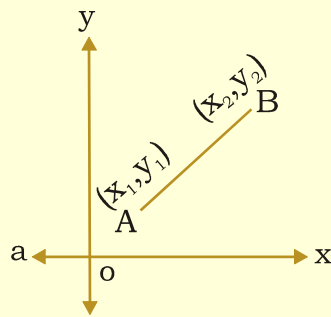
$$\therefore AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

NOTE : The distance to the point $A(x_1, y_1)$ from origin is $\sqrt{x_1^2 + y_1^2}$

Illustration : 1. Find the distance between the points (1, 2) and (3, 2)

Solution : Let $A = (x_1, y_1) = (1, 2)$

$$B = (x_2, y_2) = (3, 2)$$



$$\therefore \text{Distance of } AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(3 - 1)^2 + (2 - 2)^2} = \sqrt{(2)^2 + (0)^2} = 2 \text{ units}$$

Illustration : 2. Find the distance to the point (2,1) from origin

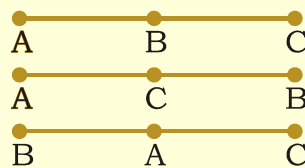
Solution : Let $A = (x_1, y_1) = (2, 1)$

$$\therefore \text{Distance of } OA = \sqrt{x_1^2 + y_1^2} = \sqrt{2^2 + 1^2} = \sqrt{5} \text{ units}$$

Collinear points : The points which lie on the same line are called collinear points, If three points A, B, C are collinear then $AB + BC = AC$

$$(\text{or}) AC + CB = AB$$

$$(\text{or}) BA + AC = BC$$



Note 1) : If A,B,C are collinear then A,B,C lie on the same line

Note 2) : A,B,C are collinear, then the area of $\triangle ABC$ is zero

Illustration : Show that the points (-1, 7), (3, -5), (4, -8) are collinear

Solution : Let $A = (-1, 7)$, $B = (3, -5)$, $C = (4, -8)$ are given points

the distance between two points $(x_1, y_1), (x_2, y_2)$ is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

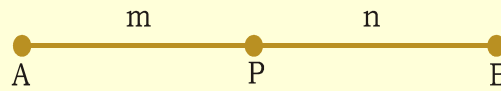
$$\therefore AB = \sqrt{(3 - (-1))^2 + (-5 - 7)^2} = \sqrt{4^2 + (-12)^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10} \text{ units}$$

$$BC = \sqrt{(4 - 3)^2 + (-8 - (-5))^2} = \sqrt{1^2 + 3^2} = \sqrt{10} \text{ units}$$

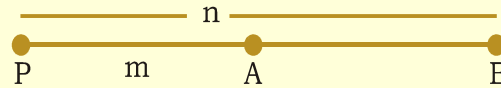
$$CA = \sqrt{(-1 - 4)^2 + (7 - (-8))^2} = \sqrt{25 + 225} = \sqrt{250} = 5\sqrt{10} \text{ units}$$

$$\text{Now, } AB + BC = 4\sqrt{10} + \sqrt{10} = 5\sqrt{10} = AC$$

$\therefore A, B, C$ are collinear.

Dividing a line segment in a given ratio (section formulae) :

(P divides $\overline{AB}$ in the ratio $m : n$ internally.)



(P divides $\overline{AB}$ in the ratio $m : n$ externally.)

Section formulae :

The point 'P' which divides the line segment joining the points $A(x_1, y_1), B(x_2, y_2)$ in the ratio $m:n$

i) internally is $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right); m+n \neq 0$

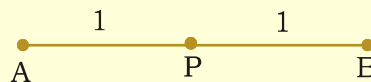
ii) externally is $\left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right); m \neq n$

Illustration : Find the points which divide the line segment joining the points $(1, -3), (-3, 9)$ in the ratio $1:2$ internally and externally.

Solution : Let $A = (x_1, y_1) = (1, -3), B = (x_2, y_2) = (-3, 9)$ and ratio $= m : n = 1 : 2$

$$\begin{aligned} \text{The point which divides } \overline{AB} \text{ in the ratio } 1:2 \text{ internally is } &= \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right) \\ &= \left(\frac{1 \times -3 + 2 \times 1}{1+2}, \frac{1 \times 9 + 2 \times -3}{1+2} \right) = \left(\frac{-1}{3}, 1 \right) \end{aligned}$$

$$\begin{aligned} \text{The point which divides } \overline{AB} \text{ in the ratio } 1:2 \text{ externally is } &= \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right) \\ &= \left(\frac{1 \times -3 - 2 \times 1}{1-2}, \frac{1 \times 9 - 2 \times -3}{1-2} \right) = \left(\frac{-3-2}{-1}, \frac{9+6}{-1} \right) = \left(\frac{5}{1}, \frac{-15}{1} \right) = (5, -15) \end{aligned}$$

Mid point of a line segment :

Let $A = (x_1, y_1), B = (x_2, y_2)$

If P is the mid point of the line segment $\overline{AB}$ then P divides $\overline{AB}$ in the ratio $1:1$ internally. Let $m:n=1:1$

$$\text{Now } P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right) = \left(\frac{1 \times x_2 + 1 \times x_1}{1+1}, \frac{1 \times y_2 + 1 \times y_1}{1+1} \right) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

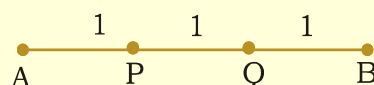
i.e., the mid point of line segment joining of (x_1, y_1) and (x_2, y_2) is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Illustration : Find the mid point of the line segment joining (1, 2) and (1, 4).

Solution : Let $A = (x_1, y_1) = (1, 2)$, $B = (x_2, y_2) = (1, 4)$

$$\text{Now, mid point of } \overline{AB} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = \left(\frac{1+1}{2}, \frac{2+4}{2} \right) = (1, 3)$$

The points of trisection of a line segment :



P divides $\overline{AB}$ in the ratio 1:2; Q divides $\overline{AB}$ in the ratio 2:1

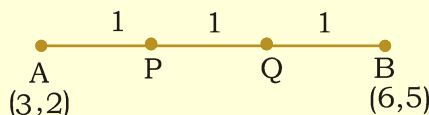
Here P, Q are called the **points of trisection** of the line segment $\overline{AB}$

Note : The points which divide a line segment in the ratio 1:2 and 2:1 are called the points of trisection of the line segment.

Illustration : Find the points of trisection of the segment joining (3, 2) and (6, 5)

Solution : Let $A = (x_1, y_1) = (3, 2)$, $B = (x_2, y_2) = (6, 5)$

Let the point P divides $\overline{AB}$ in the ratio 1:2 and the point Q divides $\overline{AB}$ in the ratio 2:1, internally.



When $m : n = 1 : 2$

$$\therefore P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right) = \left(\frac{1(6) + 2(3)}{1+2}, \frac{1(5) + 2(2)}{1+2} \right) = (4, 3)$$

When $m : n = 2 : 1$

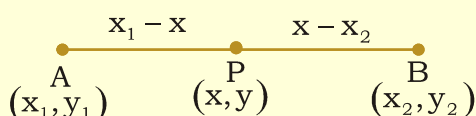
$$\therefore Q = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right) = \left(\frac{2(6) + 1(3)}{2+1}, \frac{2(5) + 1(2)}{2+1} \right) = (5, 4)$$

$\therefore$ The points of trisection are (4,3), (5,4).

NOTE :

1. The point $P(x, y)$ divides the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ in

$$x_1 - x : x - x_2 \text{ (or) } y_1 - y : y - y_2$$



$$\text{i.e. } AP = PB = x_1 - x = x - x_2$$

2. x-axis divides the line segment joining (x_1, y_1) and (x_2, y_2) in the ratio $-y_1 : y_2$
3. y-axis divides the line segment joining (x_1, y_1) and (x_2, y_2) in the ratio $-x_1 : x_2$

Illustration :

1. Find the ratio in which (2, 1) divides the line segment joining (1, 4) & (4, 5).

Solution : Let $P(x, y) = (2, 1)$, given $A(x_1, y_1) = (1, 4)$ and $B(x_2, y_2) = (4, 5)$

$$\therefore \text{The required ratio} = x_1 - x : x - x_2 = 1 - 2 : 2 - 4 = -1 : -2 = 1 : 2$$

$\therefore$ The points divides the line segment in the ratio of 1:2 internally

Illustration :

2. Find the ratio in which x-axis and y-axis divide the line segment joining the points $(-3, 5)$ and $(3, -7)$.

Solution : Let $A(x_1, y_1) = (-3, 5)$ and $B(x_2, y_2) = (3, -7)$

The ratio in which

$$\therefore \text{x-axis divides } \overline{AB} = -y_1 : y_2 = -5 : -7 = 5 : 7$$

The ratio in which

$$\therefore \text{y-axis divides } \overline{AB} = -x_1 : x_2 = -(-3) : 3 = 1 : 1$$

Second - order determinant :

The expression $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is called the second-order determinant.

It is defined as $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

$$\text{Example : } \begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix} = (4)(1) - 3(2) = 4 - 6 = -2.$$

Area of a triangle :

1. The area of the triangle formed by the points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$

$$= \frac{1}{2} \left| \sum x_i (y_j - y_k) \right| \quad (\text{or}) \quad \frac{1}{2} \begin{vmatrix} x_1 - x_2 & x_1 - x_3 \\ y_1 - y_2 & y_1 - y_3 \end{vmatrix} \quad (\text{or}) \quad \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix} \text{ sq.units}$$

2. The area of the triangle formed by the points $O(0, 0)$, $A(x_1, y_1)$, $B(x_2, y_2)$

$$= \frac{1}{2} |x_1 y_2 - x_2 y_1| \text{ sq.units.}$$

Illustration : 1. Find the area of the triangle formed by the points (5,1), (9,3) and (3,5)

Solution : Let $A = (x_1, y_1) = (5, 2)$

$$B = (x_2, y_2) = (9, 3)$$

$$C = (x_3, y_3) = (3, 5)$$

$$\begin{aligned} \therefore \text{Area of } \triangle ABC &= \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix} = \frac{1}{2} \begin{vmatrix} 5 & 9 & 3 & 5 \\ 2 & 3 & 5 & 2 \end{vmatrix} \\ &= \frac{1}{2} |15 - 18 + 45 - 9 + 6 - 25| = \frac{1}{2} |-3 + 36 - 19| = \frac{1}{2} |14| = 7 \text{ sq. units.} \end{aligned}$$

Illustration : Find the area of the triangle formed by the points (0,0), (1,0) and (0,1).

Solution: Let O=(0,0)

$$A = (x_1, y_1) = (1, 0)$$

$$B = (x_2, y_2) = (0, 1)$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{2} |x_1 y_2 - x_2 y_1| = \frac{1}{2} |(1)(1) - (0)(0)| = \frac{1}{2} |1 - 0| = \frac{1}{2} \text{ sq unit.}$$

NOTE :

1. Three points A,B,C are collinear if the area of $\triangle ABC$ is zero.
2. If D,E,F are the mid points of the sides of the $\triangle ABC$ then the area of $\triangle DEF = \frac{1}{4}$ (area of $\triangle ABC$).
3. If G is the centroid of the $\triangle ABC$ then area of $\triangle ABC = 3$ (area of $\triangle GAB$)

Worksheet - 1, 2

Date: 16-08-2010

LEVEL - 1

SINGLE CORRECT CHOICE TYPE:

1. Let A(a cos θ , 0), B(0, a sin θ) be any two points then the distance between two points is
 - 1) a units
 - 2) a^2 units
 - 3) $\sqrt{a}$ units
 - 4) $\sqrt{2}a$ units
2. If the points (3,-8), (4,-11) and (5,-k) are collinear then, the value of k is
 - 1) 14
 - 2) -8
 - 3) 4
 - 4) 5
3. The triangle formed by (0,1), (1,0) and (1,1) is
 - 1) Right angle isosceles triangle
 - 2) Scalene triangle
 - 3) Equilateral triangle
 - 4) Cannot form a triangle
4. The mid point of the line joining the points (1,4) and (x,y) is (2,3) then (x + y) is
 - 1) 5
 - 2) $\frac{5}{2}$
 - 3) 7
 - 4) -5

5. If the point $p(2,3)$ divides the line joining the points $(5,6)$ and $(8,9)$, then the ratio is

- 1) 1:2 internally 2) 1:2 externally 3) 2:1 internally 4) 2:1 externally

MULTI CORRECT CHOICE TYPE:

6. Let $A(x_1, y_1)B(x_2, y_2)$ be any two points distance between two points

- 1) $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ 2) $\sqrt{(x_2 + x_1)^2 + (y_2 - y_1)^2}$
 3) $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ 4) $\sqrt{(x_2 + x_1)^2 + (y_2 - y_1)^2}$

REASONING TYPE:

7. *Statement I* : The points $A(5,2), B(7,9), C(9,16)$ are collinear.

Statement II : If three points A, B, C are collinear then $AB + AC = BC$ or $AC + CB = AB$ or $AB + BC = AC$.

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

COMPREHENSION TYPE:

The points A, B and C are called collinear points if they lie on the same line. If three points are collinear then $AB + BC = AC$ (or) $AC + CB = AB$ (or) $AC + AB = BC$.

Based on the above passage answer the following questions.

8. If $A(2, 7), B(8, 3)$ and $C(14, -1)$ are collinear then which of the following is true.

- 1) $AB + BC = AC$ 2) $AC + CB = AB$ 3) $AB + AC = BC$ 4) All the above

9. If A, B and C are collinear then the area of $\triangle ABC$ is

- 1) Positive real 2) Negative real 3) 0 4) Complex number

10. If $A(-1, 3), B(2, p)$ and $C(5, -1)$ are collinear then the value of p is

- 1) 0 2) 3 3) 2 4) 1

MATRIX MATCH TYPE:

11. **Column-I**

a) Distance between $(5, 3), (8, 7)$

b) Distance between $(0, 0), (-4, 3)$

c) Distance between $\left(\frac{1}{2}, \frac{3}{2}\right), \left(\frac{3}{2}, \frac{-1}{2}\right)$

d) If $(1, x)$ is at $\sqrt{10}$ units from $(0, 0)$ then $x =$

Column-II

1) -3

2) $\sqrt{5}$

3) 3

4) 5

5) 4

INTEGER ANSWER TYPE:

12. $A(0, 2), B(0, 5)$ be any two points, then the distance between them is _____

LEVEL - 2**SINGLE CORRECT CHOICE TYPE:**

13. The coordinates of the point which divides the line segment joining points A (0,0) and B(9, 12) in the ratio 1 : 2 are

1) (-3, 4) 2) (3, 4) 3) (3, -4) 4) None of these

MULTI CORRECT CHOICE TYPE:

14. ABC is an isosceles triangle whose base is BC. If B and C are $(a + b, b - a)$ and $(a - b, a + b)$ then coordination of A may be

1) (a,b) 2) (b,a) 3) $\left(\frac{a}{b}, \frac{b}{a}\right)$ 4) $\left(1, \frac{b}{a}\right)$

LEVEL - 3**SINGLE CORRECT CHOICE TYPE:**

15. The point which divides the line joining the points $(a + b, a - b)$ and $(a - b, a + b)$ in the ratio $a : b$ is

1) $\left(\frac{a^2 + b^2}{a + b}, \frac{(a + b)^2}{a + b}\right)$ 2) $\left(\frac{a^2 + b^2}{a + b}, \frac{b^2 + ab}{a + b}\right)$
 3) $\left(\frac{a^2 + b^2}{a + b}, \frac{a^2 - b^2 + 2ab}{a + b}\right)$ 4) None of these

16. The ratio in which the line segment joining the points (3, -4) and (-5, 6) is divided by the x - axis is

1) 2 : 3 2) 3 : 2 3) -2 : 3 4) None of these

REASONING TYPE:

17. *Statement I* : The vertices (3, 2), (0, 5), (-3, 2), (0, -1) represents the vertices of a square.

Statement II : The sides of a square are equal and the diagonals of it bisect each other and are equal.

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

18. Let P and Q be the points on the line segment joining A(-2, 5) and B(3, 1) such that $AP = PQ = QB$. Then the midpoint of PQ is

1) $\left(\frac{1}{2}, 3\right)$ 2) $\left(-\frac{1}{2}, 4\right)$ 3) (2,3) 4) (-1,4)

MATRIX MATCH TYPE:

19. If $(p + 1, 1)$, $(2p + 1, 3)$ and $(2p + 2, 2p)$ are the vertices of a triangle then match the following.

Column-I**Value of p**

- a) $p = 0$
 b) $p = \pm 1$
 c) $p = 3$
 d) $p = -3$

Column-II**Area of the triangle**

- 1) $7/2$
 2) $25/2$
 3) $3/2$
 4) 1

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

20. The point whose coordinates are $x = x_1 + t(x_2 - x_1)$, $y = y_1 + t(y_2 - y_1)$ divides the join of (x_1, y_1) and (x_2, y_2) in the ratio

- 1) $\frac{t}{1+t}$ 2) $\frac{1+t}{t}$ 3) $\frac{t}{1-t}$ 4) $\frac{1-t}{t}$

21. If the point $(x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1))$ divides the join of (x_1, y_1) and (x_2, y_2) internally, then

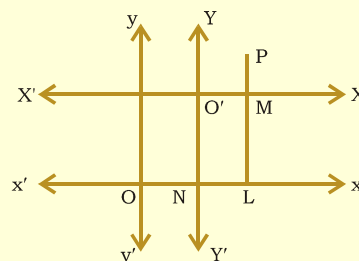
- 1) $t < 0$ 2) $t = 1$ 3) $0 < t < 1$ 4) None of these

Synopsis**Date: 17-08-2010****TRANSFORMATION OF AXES**

Definition : If the origin is shifted to another point without changing the directions of the axes, then that transformation is called **Transformation of axes**.

1. If the coordinates (x, y) of a point are transformed to (X, Y) , when the axes are translated to the point (h, k) , then $x = X + h$, $y = Y + k$.

Proof : Let $O'(h, k)$ be the point to which the origin is shifted. Let $x'Ox$, $y'Oy'$ be the original axes. Let $X'O'X$, $Y'O'Y'$ be the new axes. Then $x'Ox$, $X'O'X$ are parallel. Let P be the given point whose coordinates are (x, y) with reference to the original axes and (X, Y) with reference to the new axes. Let L, M be the projections of P on $\overline{x'Ox}$ and $\overline{X'O'X}$ respectively.



Let N be the projection of O' on $\overline{x'Ox}$. Then $x = OL = ON + NL = ON + O'M$
 $= h + X = X + h$
 $y = PL = PM + ML = PM + O'N = Y + k$.

2. If the axes are translated to the point (h, k) then the equation $f(x, y) = 0$ of a curve is transformed to $f(X + h, Y + k) = 0$.
3. If $f(X, Y) = 0$ is the transformed equation of a curve when the axes are translated to the point (h, k) , then the original equation of the curve is $f(x - h, y - k) = 0$.
4. If the axes are translated to the (h, k) then the equation $f(x, y) = 0$ of a curve is transformed to $f(X + h, Y + k) = 0$.
5. If $f(X, Y) = 0$ is the transformed equation of a curve when the axes are translated to the point (h, k) , then the original equation of the curve is $f(x - h, y - k) = 0$.
6. The point to which the origin has to be shifted to eliminate x, y terms in

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is } \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right) \text{ where } (h^2 \neq ab)$$

7. The point to which the origin has to be shifted to eliminate x and y terms in

$$ax^2 + by^2 + 2gx + 2fy + c = 0 \text{ is } \left(\frac{-g}{a}, \frac{-f}{b} \right).$$

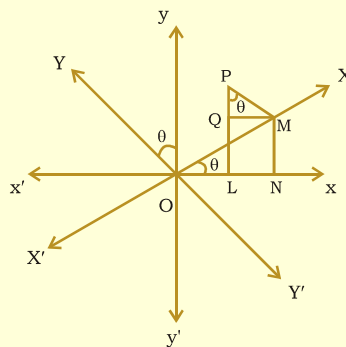
ROTATION OF AXES :

Definition : If the axes are rotated through an angle θ about the origin then that transformation is called **rotation of axes**.

1. If the coordinates (x, y) of a point are transformed to (X, Y) , when the axes are rotated through an angle θ about the origin then

$$x = X \cos \theta - Y \sin \theta, y = X \sin \theta + Y \cos \theta.$$

Proof : Let $x'Ox, y'Oy'$ be the original axes. Let $X'OX, Y'OY'$ be the new axes. Then $\angle xOx' = \theta$, $\angle yOy' = \theta$. Let P be the given point whose coordinates are (x, y) with reference to the original axes and (X, Y) with reference to the new axes.



Let L, M be the projections of P on $\overline{x'Ox}$ and $\overline{X'OX}$ respectively. Let N, Q be the projections of M on $\overline{x'Ox}$ and $\overline{Y'OY'}$ respectively. Now $\angle QPM = \theta$.

$$\text{From } \triangle OMN, \cos \theta = \frac{ON}{OM}, \sin \theta = \frac{MN}{OM} \Rightarrow ON = OM \cos \theta, MN = OM \sin \theta$$

$$\text{From } \triangle PQM, \cos \theta = \frac{PQ}{PM}, \sin \theta = \frac{QM}{PM} \Rightarrow PQ = PM \cos \theta, QM = PM \sin \theta$$

$$x = OL = ON - LN = ON - QM = OM \cos \theta - PM \sin \theta = X \cos \theta - Y \sin \theta$$

$$y = PL = PQ + QL = PQ + MN = PM \cos \theta + OM \sin \theta = Y \cos \theta + X \sin \theta = X \sin \theta + Y \cos \theta$$

$$\therefore x = X \cos \theta - Y \sin \theta, y = X \sin \theta + Y \cos \theta$$

By solving the above, we get $\Rightarrow x = x \cos \theta + y \sin \theta, \Rightarrow Y = -x \sin \theta + y \cos \theta$

2. The transformation of coordinates when the axes are rotated through angle θ can be obtained from the following **transformation table**.

θ	X	Y
x	$\cos \theta$	$-\sin \theta$
y	$\sin \theta$	$\cos \theta$

Worksheet - 3

Date: 17-08-2010

LEVEL - 1

SINGLE CORRECT CHOICE TYPE:

- The point to which the axes be translated so as to remove the first degree terms in the equation $2x^2 + 3xy - 2y^2 - 7x + y - 2 = 0$ is
 1) $(-1, 1)$ 2) $(1, 1)$ 3) $(-1, -1)$ 4) $(1, -1)$
- The new origin so that the equation $2xy + 2x + 2y + 1 = 0$ may be translated to the form in which the first degree terms be missing, is
 1) $(2, 2)$ 2) $(-1, -1)$ 3) $(1, 1)$ 4) $(-2, -2)$
- The first degree terms of the equation $x^2 + y^2 + 2x - 4y + 3 = 0$ be removed by shifting to a new origin given by
 1) $(1, -2)$ 2) $(-1, 2)$ 3) $(2, 1)$ 4) $(-2, 1)$

MULTI CORRECT CHOICE TYPE:

- The angle of rotation of axes to remove xy term in the equation $9x^2 - 2\sqrt{3}xy + 3y^2 = 0$ is
 1) $\pi/12$ 2) $\frac{1}{2}\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$ 3) $\frac{1}{2}\tan^{-1}(1)$ 4) $\frac{5\pi}{12}$

REASONING TYPE:

5. *Statement I* : The angle of rotation to remove the xy term in the equation

$$2x^2 + \sqrt{3}xy + 3y^2 = 9 \text{ is } \pi/6$$

Statement II : The angle of rotation of the axes to eliminate xy term in the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is } \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$$

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

COMPREHENSION TYPE:

The point to which the origin has to be shifted to eliminate x,y terms in

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is } \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$$

Based on the above passage answer the following questions.

6. The point to which the axes are to be translated to eliminate x and y terms in the equation $3x^2 - 4xy - 2y^2 - 3x - 2y - 1 = 0$ is

- 1) (-2, 1) 2) (-4, 3) 3) (-2, 3) 4) $\left(\frac{1}{10}, -\frac{3}{5}\right)$

7. The point to which the origin should be shifted in order to eliminate x and y terms in the equation $x^2 - y^2 + 2x + 4y = 0$ is

- 1) (1,3) 2) (-4, 3) 3) (-1,2) 4) (1,-2)

8. The point to which the origin should be shifted in order to eliminate x and y terms in the equation $x^2 + y^2 - 2ax - 4ay + a^2 = 0$ is

- 1) (2,a) 2) (5,-7) 3) (2,-1) 4) (a,2a)

MATRIX MATCH TYPE:

9. Let us suppose that origin is shifted to (1,2) then match the following

Column-I

- a) (7,5) changes to
b) (-3,1) changes to
c) (0,5) change to
d) (-1,-2) changes to

Column-II

- 1) (-4,-1)
2) (-2,-4)
3) (6,3)
4) (-1,3)

INTEGER ANSWER TYPE:

10. The origin is shifted to (2, -3). The equation of the curve $y^2 - 4x + 6y + 17 = 0$ has changed to $y^2 = 4ax$. Then a = _____

LEVEL - 2**SINGLE CORRECT CHOICE TYPE:**

11. By translating the axes the equation $xy - x + 2y = 6$ has changed to $xy = c$, then c =
1) 4 2) 5 3) 6 4) 7
12. The origin is shifted to (-2, 1). Then the equation $2x^2 + 7y^2 + 8x - 14y + 15 = 0$ changes to
1) $2x^2 + 7y^2 = 10$ 2) $2x^2 + 7y^2 = 0$ 3) $2x^2 + 7y^2 = 2$ 4) none

REASONING TYPE:

13. *Statement I* : If the axes rotated through an angle 30° in the clockwise direction, the coordinates of $(4, 2\sqrt{3})$ before the rotation of axes are $(\sqrt{3}, 5)$

Statement II : When the axes are rotated through an angle θ the relation between (x,y) and (x',y') is $x = x' \cos \theta - y' \sin \theta$, $y = x' \sin \theta + y' \cos \theta$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

LEVEL - 3**SINGLE CORRECT CHOICE TYPE:**

14. The transformed equation of $2x^2 + 4xy + 5y^2 - 4x - 22y + 7 = 0$ when the axes are translated to the point $(-2, 3)$ is
- 1) $2x^2 + 4xy + 5y^2 = 11$
 - 2) $2x^2 + 4xy + 5y^2 = 33$
 - 3) $2x^2 + 4xy + 5y^2 = 22$
 - 4) none
15. The transformed equation $4x^2 + 9y^2 - 8x + 36y + 4 = 0$ when the axes are translated to the point $(1, -2)$ is
- 1) $4x^2 + 9y^2 = 16$
 - 2) $4x^2 + 9y^2 = 26$
 - 3) $4x^2 + 9y^2 = 36$
 - 4) none

REASONING TYPE:

16. *Statement I* : The point $\left(\frac{-1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ is transformed to $(4, -3)$ when axes are rotated through an angle of 135° .

Statement II : If the axes are rotated through an angle ' θ ' then the equation

$f(x, y) = 0$ of a curve is transformed to $f(x \cos \theta - y \sin \theta, x \sin \theta + y \cos \theta) = 0$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

17. When the axes are translated to the point $(2, -2)$ the equation $2x^2 + y^2 - 8x + 4y + 1 = 0$ changes to
- 1) $2x^2 - y^2 = 1$
 - 2) $2x^2 + y^2 = 10$
 - 3) $2x^2 + y^2 = 11$
 - 4) none

MATRIX MATCH TYPE:

18. To remove the first degree terms in the following equations origin should be shifted to the another point then calculate the new origins for

Column-I**Column-II**

- | | |
|-------------------------------------|------------------------------------|
| a) $x^2 - y^2 + 2x + 4y = 0$ | 1) $\left(5, \frac{-26}{3}\right)$ |
| b) $4x^2 + 9y^2 - 8x + 36y + 4 = 0$ | 2) $(1, -2)$ |
| c) $4x^2 + 9y^2 - 8x + 36y + 4 = 0$ | 3) $(-1, 2)$ |
| d) $2(x - 5)^2 + 3(y + 7)^2 = 10$ | 4) $(-1, -2)$ |
| | 5) $(-5, 7)$ |

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

19. The origin is shifted to (1, 2). The equation $y^2 - 8x - 4y + 12 = 0$ change to $y^2 = 4ax$ then a is equal to
 1) 1 2) 2 3) -2 4) -1
20. Origin is shifted to the point (1, 1). If $(\cos \alpha, \cos \beta)$ are the new coordinates of the point P, then the original point P =
 1) $\left(2\cos^2 \frac{\alpha}{2}, 2\cos^2 \frac{\beta}{2}\right)$ 2) $\left(2\sin^2 \frac{\alpha}{2}, 2\sin^2 \frac{\beta}{2}\right)$
 3) $\left(2\cos \frac{\alpha}{2}, 2\cos \frac{\beta}{2}\right)$ 4) $\left(2\sin \frac{\alpha}{2}, 2\sin \frac{\beta}{2}\right)$

MATRIX MATCH TYPE:**21. Column-I****Column-II**

- a) If the axes are rotated through an angle 18° the transformed equation of $x^2 + y^2 = 25$ is
 1) $x^2 + y^2 = 9$
- b) The transformed equation of $17x^2 - 16xy + 17y^2 = 225$
 2) $x^2 + y^2 = 25$
- c) If the axes are rotated through an angle 60° then transformed equation of $x^2 + y^2 = 9$
 3) $x^2 - y^2 = -1$
- d) If the axes are rotated through an angle 90° then the transformed equation of $x^2 - y^2 = 1$ is
 4) $17x^2 + 25y^2 = 225$

Cumulative Assignment**Date: 18-08-2010****LEVEL - 1****SINGLE CORRECT CHOICE TYPE:**

1. The point to which the origin be shifted in order to make the first degree terms missing in the equation $2x^2 + 7y^2 + 8x - 14y + 15 = 0$ is
 1) (-2, 1) 2) (1, 2) 3) (2, 1) 4) (1, -2)
2. Without changing the direction of the axes, origin is transferred to (h, k), so that the linear terms in the equation $x^2 + y^2 - 4x + 6y - 7 = 0$ are eliminated. Then (h, k) =
 1) (3, 2) 2) (2, -3) 3) (-3, 2) 4) (2, -3)

3. If the axes are rotated through an angle of 45° , then the coordinates of $(\sqrt{2}, -\sqrt{2})$ in the new system are
- 1) (2,0) 2) (0,2) 3) (-2,0) 4) (0,-2)

MULTI CORRECT CHOICE TYPE:

4. The transformed equation of $17x^2 - 16xy + 17y^2 = 225$ when the axes are rotated through an angle 45° is

- 1) $9x^2 + 25y^2 = 225$ 2) $\frac{x^2}{25} + \frac{y^2}{9} = 1$
- 3) $9x^2 - 25y^2 = 225$ 4) $25x^2 + 9y^2 = 225$

REASONING TYPE:

5. *Statement I* : The point to which the origin should be shifted in order to eliminate x and y terms in the equation $2x^2 - 3y^2 - 12x - 6y + 5 = 0$ is (3, -1)

Statement II : The point to which the origin has to be shifted to eliminate x and

y terms in $ax^2 + by^2 + 2gx + 2fy + c = 0$ is $\left(\frac{-g}{a}, \frac{-f}{b}\right)$

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

COMPREHENSION TYPE:

If the coordinates (x,y) of a point are transformed to (X,Y), when the axes are rotate through an angle θ about the origin then

$$x = X \cos \theta - Y \sin \theta, y = X \sin \theta + Y \cos \theta.$$

6. The transformed equation of $7x^2 + 2\sqrt{3}xy + 9y^2 = 8$ when the axes are rotated through an angle $\frac{\pi}{3}$ is
- 1) $3x^2 - y^2 + 2\sqrt{2}y - 6 = 0$ 2) $5x^2 + 3y^2 = 5$
- 3) $5x^2 + 3y^2 = 4$ 4) $4x + 3y^2 = 6$
7. The transformed equation of $x \sin \alpha - y \cos \alpha = p$ when the axes are rotated through an angle α is
- 1) $x = p$ 2) $y = p$ 3) $x + p = 0$ 4) $y + p = 0$
8. The transformed equation of $x^2 + 4xy + y^2 - 2x + 2y - 6 = 0$ when the axes are rotated through an angle $\frac{\pi}{4}$ is
- 1) $3x^2 - y^2 + 2\sqrt{2}y - 6 = 0$ 2) $5x^2 + 3y^2 = 5$
- 3) $5x^2 + 3y^2 = 4$ 4) $4x^2 + 3y^2 = 6$

MATRIX MATCH TYPE:**9. Column-I****Column-II**

- a) The transformed equation of $x^2 - y^2 + 2x + 4y = 0$ When the origin is shifted to the point $(-1, 2)$ 1) $x^2 + 3y^2 = 1$
- b) The transformed equation of $x^2 + 3y^2 + 4x + 18y + 30 = 0$ when the axes are translated to $(-2, -3)$ is 2) $4x^2 + 9y^2 = 36$
- c) The transformed equation of $4x^2 + 9y^2 - 8x + 36y + 4 = 0$ when axes are translated to the point $(1, -2)$ 3) $4x^2 + 9y^2 = 36$
- d) The transformed equation of $2x^2 + 4xy + 5y^2 - 22y + 7 = 0$ when the axes are translated to $(-2, 3)$ is 4) $2x^2 + 4xy + 5y^2 = 22$

LEVEL - 2**SINGLE CORRECT CHOICE TYPE:**

10. If the axes are rotated through an angle of 60° , the coordinates of a point in the new system are $(2, -\sqrt{3})$ then its original coordinates are
 1) $(5/3 - \sqrt{2}/3)$ 2) $(-5/3, \sqrt{2}/3)$ 3) $(5/2, \sqrt{3}/2)$ 4) $(-5/2, \sqrt{3}/2)$
11. The y term in the equation $x\sqrt{3} + y = 2$ is removed by suitable rotation of axes. The angle of rotation is
 1) 90° 2) 60° 3) 45° 4) 30°

LEVEL - 3**SINGLE CORRECT CHOICE TYPE:**

12. By rotating axes through an angle α , the equation $x \cos \alpha + y \sin \alpha = p$ changes to
 1) $x = p$ 2) $y = p$ 3) $x + y = p$ 4) none

REASONING TYPE:

13. *Statement I*: If the transformed equation of a curve is $9x^2 + 16y^2 = 144$ when the axes are rotated through an angle 45° , then the original equation is $25x^2 - 14xy + 25y^2 = 288$.
- Statement II*: If $f(x, y) = 0$ is the transformed equation of a curve when the axes are rotated through an angle θ then the original equation of the curve is $f(x \cos \theta + y \sin \theta, -x \sin \theta + y \cos \theta) = 0$.
- Both Statements are true, Statement II is the correct explanation of Statement I.
 - Both Statements are true, Statement II is not correct explanation of Statement I.
 - Statement I is true, Statement II is false.
 - Statement I is false, Statement II is true.

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

14. By rotating the axes through 180° , the equation $x - 2y + 3 = 0$ change to
 1) $x + 2y - 3 = 0$ 2) $x - 2y + 3 = 0$ 3) $x - 2y - 3 = 0$ 4) none
15. If the axes are rotated through an angle of 90° in anticlockwise direction then new equation of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
 1) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ 2) $\frac{x^2}{b^2} - \frac{y^2}{a^2} = 1$ 3) $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ 4) none

REASONING TYPE:

16. Statement - I : If the axes are rotated through an angle 30° in the clockwise direction, the coordinates of $(4, 2\sqrt{3})$ before the rotation of axes are $(\sqrt{3}, 5)$.

Statement - II: When the axes are rotated through an angle θ the relation between (x, y) and (X, Y) is $x = X \cos \theta - Y \sin \theta, y = X \sin \theta + Y \cos \theta$

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

17. If the axes are rotated through an angle of 45° in clockwise direction then new equation of $x^2 - y^2 = a^2$ is
 1) $xy = a^2$ 2) $xy = 2a^2$ 3) $2xy = a^2$ 4) $2xy + a^2 = 0$
18. Let A be the image of $(2, -1)$ w.r.t. to y - axis. Without transforming the origin, the axes are turned through an angle of 45° in the clockwise direction . Then A in new system is
 1) $\left(\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$ 2) $\left(\frac{-1}{\sqrt{2}}, -\frac{3}{\sqrt{2}}\right)$ 3) $\left(\frac{-3}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ 4) none

MATRIX MATCH TYPE:

19. The angle of rotation of axes to remove the xy term

Column-I**Column-II**

a) $9x^2 + 2\sqrt{3}xy + 7y^2 = 0$

1) $\frac{\pi}{2}$

b) $7x^2 + 2\sqrt{3}xy + 9y^2 = 0$

2) $\frac{\pi}{4}$

c) $3x^2 + 2xy + 3y^2 = 2$

3) $\frac{\pi}{3}$

d) $3x^2 - 2\sqrt{3}xy + 9y^2 = 10$

4) $\frac{\pi}{6}$

5) $-\frac{\pi}{6}$

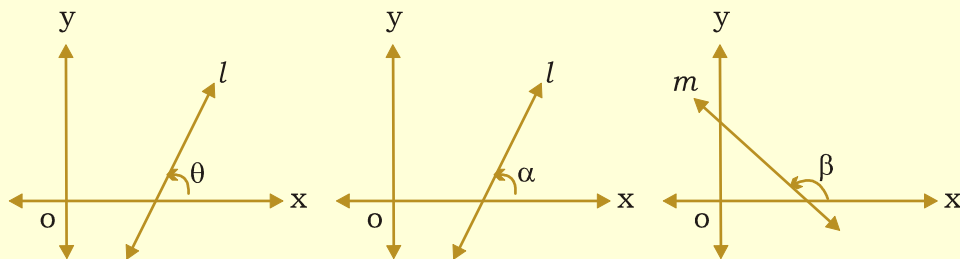
Synopsis

Date: 19-08-2010

Inclination of a line : The angle made by a line with x-axis in the anticlockwise direction is called its steepness or inclination.

If θ is the inclination of the line, then $0^\circ \leq \theta < 180^\circ$

The following lines l , m are making angles α, β respectively with x-axis



Slope of a line :

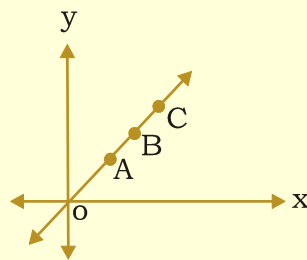
If ' θ ' is the inclination of the line, then $\tan \theta$ is called the slope of the line. It is denoted by 'm' i.e., $m = \tan \theta$ or

The ratio between the difference of y co-ordinates and x co-ordinates of any two points on the line is a constant, this constant ratio is called the slope of the given line.

Illustration :

Let us find the slope of the line $y = x$.

Let O, A, B, C, be the points on the line with O = (0, 0), A = (1, 1), B = (2, 2) and C(3, 3). Now, for any two points of O, A, B and C



$$\frac{\text{Difference of y-co-ordinates}}{\text{Difference of x-co-ordinates}} = \frac{1-0}{1-0} = \frac{2-1}{2-1} = \frac{3-2}{3-2} = 1 \text{ a constant.}$$

This constant is called the slope of the given line $\therefore m = 1$

Formula :

If ' θ ' is the inclination of a line and $(x_1, y_1), (x_2, y_2)$ are any two points on it then its

$$\text{slope } (m) = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{Difference of y co-ordinates}}{\text{Difference of x co-ordinates}}$$

Result : 1

Slope of a line parallel to x-axis is 'zero'.

We know the equation of the line parallel to x-axis is $y = k, k \in R$

Let $A(x_1, k)$ and $B(x_2, k)$ be two distinct points on it.

$$\text{Now slope (m)} = \frac{\text{Difference of y-co-ordinates}}{\text{Difference of x-co-ordinates}} = \frac{k - k}{x_2 - x_1} = \frac{0}{x_2 - x_1}; x_1 \neq x_2 \Rightarrow 0$$

$\therefore$ **Slope of any line parallel to x-axis is zero.**

In particular the slope of x-axis is zero

Result : 2

Slope of a line parallel to y-axis is 'not defined'.

Let the equation of a line parallel to y-axis be $x = k, k \in R$

Let $A(k, y_1)$ and $B(k, y_2)$ be two distinct points on it, clearly $y_1 \neq y_2$

$$\begin{aligned} \text{Now slope (m)} &= \frac{\text{Difference of y-co-ordinates}}{\text{Difference of x-co-ordinates}} = \frac{y_2 - y_1}{k - k} = \frac{y_2 - y_1}{0} \text{ where } y_1 \neq y_2 \\ &= \text{undefined } (\because \text{division with zero}) \end{aligned}$$

$\therefore$ Slope of any line parallel to y-axis is 'undefined'

In particular the slope of y-axis is 'undefined'.

NOTE : 1. If slope of $\overline{AB}$ = slope of $\overline{BC}$ then A, B, C are collinear.

2. Slope of a line $y = mx$ or $y = mx + c (m, c \in R)$ is 'm'.

Illustration : Find the slope of a line $5x + 2y = 0$

Solution : Given line is $5x + 2y = 0 \Rightarrow 2y = -5x$

$$\Rightarrow y = \frac{-5}{2}x \text{ which is of the form } y = mx \therefore \text{ slope (m)} = \frac{-5}{2}$$

Illustration : Find the slope of a line $2x + y + 7 = 0$

Solution : Given line is $2x + y + 7 = 0$

$$\begin{aligned} \Rightarrow y &= -2x - 7 \Rightarrow y = -2x + (-7), \text{ which is of the form of } y = mx + c \\ \therefore \text{ slope (m)} &= -2. \end{aligned}$$

Illustration : Find the slope of a non-vertical line $ax + by + c = 0$

Solution : Given line is $ax + by + c = 0 \Rightarrow by = -ax - c$

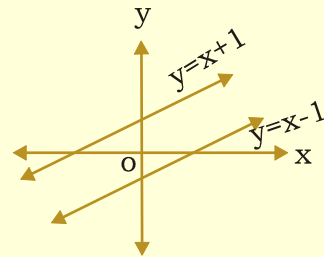
$$\Rightarrow y = \left(\frac{-a}{b}\right)x + \left(\frac{-c}{b}\right) \text{ which is in the form of } y = mx + c$$

$$\therefore \text{ slope (m)} = \frac{-a}{b}$$

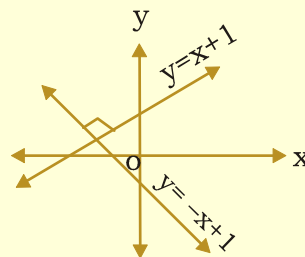
NOTE : 1) The general form of a straight line is $ax + by + c = 0$, where $|a| + |b| \neq 0$

2) Slope of a non-vertical line $ax + by + c = 0$ is $m = \frac{-a}{b} = \frac{\text{-coefficient of } x}{\text{coefficient of } y}$

By observing the following diagram



(Parallel lines)



(Perpendicular lines)

We can conclude that

1. Any two lines are parallel if their slopes are equal.
2. Any two lines are perpendicular if the product of their slopes is equal to '-1'.

NOTE : 1. Parallel lines differ only by their constants.

2. In the above two lines, no line is parallel to either of the axes.

Illustration : Show that the lines $4x + 8y = 0$, $2x + 4y = 9$ are parallel

Solution : For the given line $4x + 8y = 0$

$$\text{Slope } (m_1) = \frac{-\text{coefficient of } x}{\text{coefficient of } y} = -\frac{4}{8} = -\frac{1}{2}$$

For the given line is $2x + 4y = 9$

$$\text{Slope } (m_2) = \frac{-\text{coefficient of } x}{\text{coefficient of } y} = -\frac{2}{4} = -\frac{1}{2}$$

$$\therefore m_1 = m_2 = -\frac{1}{2} \quad \therefore \text{Given lines are parallel}$$

Illustration : Show that the lines $4x + 5y + 1 = 0$ and $5x - 4y + 7 = 0$ are perpendicular.

Solution : For the given line $4x + 5y + 1 = 0$

$$\text{Slope } (m_1) = \frac{-\text{coefficient of } x}{\text{coefficient of } y} = \frac{-4}{5}$$

For the given line $5x - 4y + 7 = 0$

$$\text{Slope } (m_2) = \frac{-\text{coefficient of } x}{\text{coefficient of } y} = \frac{-5}{-4} = \frac{5}{4}$$

$$\text{Now, } m_1 \times m_2 = \left(-\frac{4}{5}\right) \times \left(\frac{5}{4}\right) = -1$$

$\therefore$ Given lines are perpendicular

Various forms of a Straight Line**1. Slope form (Gradient form) :**

Equation of a line passing through the origin and having slope m :

Let ' l ' be the line passing through $(0, 0)$ with slope ' m '

Let $P(x, y)$ be any point on l

$$\text{Slope of } OP = \frac{y-0}{x-0} = m \Rightarrow \frac{y}{x} = m \Rightarrow y = mx$$

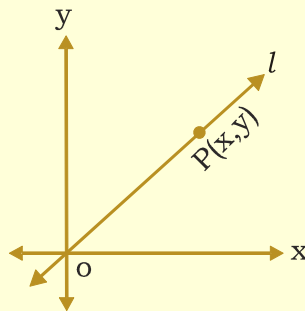
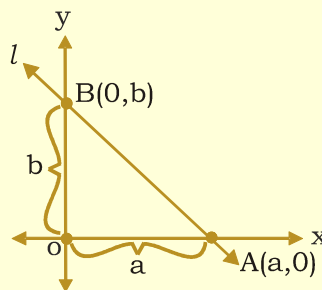


Illustration : Find the equation of the line passing through $(0, 0)$ whose slope is 5.

Solution : Slope of given line (m) = 5

Required line is $y = mx \Rightarrow y = 5x$

Intercepts of a line :

If a line ' l ' cuts x -axis at $A(a, 0)$, y -axis at $B(0, b)$ then
its x -intercept = a , y -intercept = b .

2. Slope Intercept form of a line :

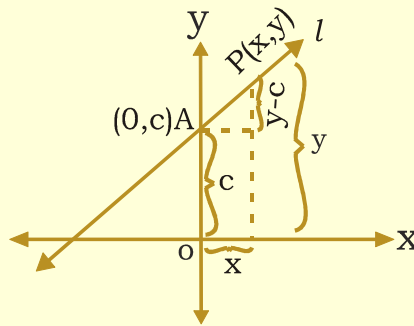
Equation of a line with slope ' m ' and the y -intercept ' c '.

If ' l ' is a line with y -intercept ' c ' and slope ' m '.

Let l meets y -axis at $A(0, c)$

Let $P(x, y)$ be a point on l

Then slope of $AP = m$



$$\Rightarrow \frac{y-c}{x-0} = m \quad \Rightarrow y-c = mx \quad \Rightarrow y = mx + c$$

Illustration : Find the equation of the line with slope 4, y -intercept 5.

Solution : Given y-intercept(c) = 5 and slope of the line (m) = 4

$\therefore$ The required equation is $y = mx + c \Rightarrow y = 4x + 5$

3. Point slope form :

The equation of a line passing through $P(x_1, y_1)$ and having slope 'm'.

If 'l' is a line passing through $P(x_1, y_1)$ and having slope m.

Let Q(x, y) be a point on l .

$$\text{Then slope of PQ} = m \Rightarrow \frac{y-y_1}{x-x_1} = m \Rightarrow y-y_1 = m(x-x_1)$$

Illustration : Find the equation of a line passing through (1, 2) having a slope 3.

Solution : Given slope of the line (m) = 3 and a point $(x_1, y_1) = (1, 2)$

$$\text{Equation of the line is } y-y_1 = m(x-x_1)$$

$$\Rightarrow y-2 = 3(x-1) \Rightarrow y-2 = 3x-3 \Rightarrow y = 3x-1$$

4. Intercept form of a line :

Let 'l' be a line with the x-intercept 'a' and y-intercept 'b'.

Then l meet the x-axis at A(a, 0) and y-axis at B(0, b). Thus A,B are two points on l.

$$\text{The slope of line } l = m = \text{slope of } AB = \frac{b-0}{0-a} = \frac{-b}{a}$$

y-intercept of l is (c) = b

$\therefore$ Equation of line in slope intercept form is $y = mx + c$

$$\Rightarrow y = \frac{-b}{a}x + b \Rightarrow bx + ay = ab, \quad \text{dividing the equation with } ab$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} = 1$$

Illustration : Find the line whose x intercept is -2 and y intercept is 3.

Solution : Given x-intercept(a) = - 2
and y-intercept(b) = 3

Equation of the line in Intercept form is $\frac{x}{a} + \frac{y}{b} = 1 \Rightarrow \frac{x}{-2} + \frac{y}{3} = 1 \Rightarrow 3x - 2y + 6 = 0$

Illustration : Find the intercepts made by $5x - 4y + 20 = 0$ on the co-ordinate axes.

Solution : Given $5x - 4y + 20 = 0$

$$\Rightarrow 5x - 4y = -20 \Rightarrow \text{x intercept} = -4 \text{ and y-intercept} = 5$$

NOTE : 1) To get x intercept substitute $y = 0$ and to get y intercept substitute $x = 0$ in the given equation i.e., $4x - 5y + 20 = 0$

$$\text{if } x = 0 \Rightarrow -5y + 20 = 0$$

$$\text{if } y = 0 \Rightarrow 4x + 20 = 0 \Rightarrow -5y = -20 \Rightarrow 4x = -20 \Rightarrow y = 4 \Rightarrow x = -5$$

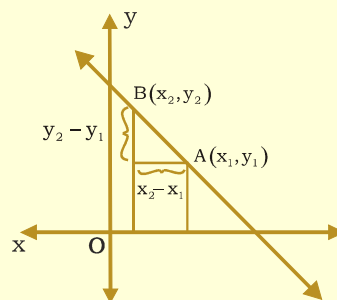
$$\therefore \text{y- intercept} = 4$$

$$\therefore \text{x-intercept} = -5$$

2) If $ax + by + c = 0$ is the equation of a line, then

$$\text{We have x intercept} = -\frac{c}{a} \text{ and y intercept} = -\frac{c}{b}$$

5. Two point form of a line :



Suppose 'l' is the line passing through the points $A(x_1, y_1), B(x_2, y_2)$.

Then the slope of 'l' is (m) = slope of AB = $\frac{y_2 - y_1}{x_2 - x_1}$

By point slope form of a line the equation of line l is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \Rightarrow (x - x_1)(y_2 - y_1) = (y - y_1)(x_2 - x_1)$$

Illustration : Find the equation of the line passing through (3,4) and (-5,1)

Solution : Given points $(x_1, y_1) = (3, 4)$ and $(x_2, y_2) = (-5, 1)$

$$\text{Equation of a line is } \frac{x - x_1}{x_1 - x_2} = \frac{y - y_1}{y_1 - y_2} \Rightarrow \frac{x - 3}{3 - (-5)} = \frac{y - 4}{4 - 1} \Rightarrow 3x - 8y + 23 = 0$$

NOTE :

- 1) For $ax + by + c = 0$ to represent a straight line the condition is $|a| + |b| \neq 0$ i.e., a & b are not simultaneously zero.
- 2) The equation of the line parallel to $ax + by + c = 0$ and passing through (x_1, y_1) is $a(x - x_1) + b(y - y_1) = 0$.
- 3) The equation of the line perpendicular to $ax + by + c = 0$ and passing through (x_1, y_1) is $b(x - x_1) - a(y - y_1) = 0$.
- 4) The equation of a line parallel to x-axis at a distance k units from it is $y = k$.
- 5) The equation of a line parallel to y-axis at a distance k units from it is $x = k$.

Worksheet - 4, 5**Date: 19-08-2010****LEVEL - 1****SINGLE CORRECT CHOICE TYPE:**

1. The equation of the line which makes 45° with x-axis and passes through $(1, 0)$ is
 1) $x + y - 1 = 0$ 2) $x - y - 1 = 0$ 3) $x + y + 1 = 0$ 4) $x - y + 1 = 0$
2. The equation of the straight line which passes through the point $(3, 4)$ and whose intercept on y - axis is twice that on x-axis, is
 1) $2x - y = 10$ 2) $x + 2y = 10$ 3) $2x + y = 10$ 4) None of these
3. The slope of the line joining the points $(-3, 2)$ and $(5, -4)$ is
 1) $-3/4$ 2) $3/4$ 3) $4/3$ 4) $-4/3$
4. The slope of the line $\frac{x}{a} + \frac{y}{b} = 1$ is
 1) $-a/b$ 2) $-b/a$ 3) a/b 4) b/a

MULTI CORRECT CHOICE TYPE:

5. If a line passes through $(2, 3)$ and $(2, -6)$ then
 1) The line is parallel to x - axis 2) The line is parallel to y - axis
 3) Slope of the line is undefined 4) Slope of the line is zero

REASONING TYPE:

6. Statement I: Equation of a line parallel to $2x + 3y + 6 = 0$ and passing through $(3, -2)$ is $2x + 3y = 0$.
 Statement II : The equation of line parallel to $ax + by + c = 0$ and passing through (x_1, y_1) is $a(x - x_1) + b(y - y_1) = 0$.
 1. Both Statements are true, Statement II is the correct explanation of Statement I.
 2. Both Statements are true, Statement II is not correct explanation of Statement I.
 3. Statement I is true, Statement II is false.
 4. Statement I is false, Statement II is true.

COMPREHENSION TYPE:

$\frac{x}{a} + \frac{y}{b} = 1$ is known as intercept form, it meet the x-axis at (a,0) and y-axis at (0,b)

7. The line $2x + 4y = 8$ meets the x -axis at
1) (4, 0) 2) (0, 2) 3) (2, 4) 4) (0, 0)
8. If a line passes through (3, 0) and (0, 4), then the equation of the line is
1) $\frac{x}{2} + \frac{y}{3} = 1$ 2) $\frac{x}{3} + \frac{y}{4} = 1$ 3) $\frac{x}{-2} - \frac{y}{-1} = 1$ 4) $\frac{x}{2} - \frac{y}{4} = 1$
9. The sum of Intercepts of a line $x + y - 10 = 0$ is
1) 30 2) 40 3) 20 4) -20

MATRIX MATCH TYPE:

- | 10. Column-I | Column-II |
|--|------------------|
| a) The slope of the line joining the the points (3, 4) and (4, 1) is | 1) 0 |
| b) The slope of the line parallel to $2x + 3y + 4 = 0$ is | 2) $-2/3$ |
| c) The slope of the line perpendicular to $x - 2y + 5 = 0$ | 3) -3 |
| d) If the line $2x - 3y = k$ passes through the origin then k is | 4) -1 |
| | 5) -2 |

INTEGER ANSWER TYRE:

11. The slope of the equations of the line $8x - 4y + 6 = 0$ is _____

LEVEL - 2

SINGLE CORRECT CHOICE TYPE:

12. The equation of the straight line which cuts off an intercept 3 from positive direction of the y-axis and is inclined at an angle 60° with the positive direction of x-axis is
- 1) $\sqrt{3}x - y - 3 = 0$ 2) $\sqrt{3}x - y + 3 = 0$ 3) $x - \sqrt{3}y - 3 = 0$ 4) $x + y\sqrt{3} = 3$

MULTI CORRECT CHOICE TYPE:

- 13(1). If $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are two perpendicular lines having slopes m_1 & m_2 then
- 1) $m_1m_2 = -1$ 2) $a_1a_2 + b_1b_2 = 0$ 3) $a_1b_2 + a_2b_1 \neq 0$ 4) $a_1a_2 + b_1b_2 = 0$
- 14(2). The ratios in which the line joining (4, 5) and (-10, -2) is divided by x axis and y axis are respectively
- 1) $1 : 2/5$, $1 : 5/2$ 2) $2 : 1$, $1 : 2$ 3) $5 : 2$, $2 : 5$ 4) $1 : 1/2$, $1/2 : 1$

LEVEL - 3

SINGLE CORRECT CHOICE TYPE:

15. Equation of the straight line joining the points $[am_1^2, 2am_1]$ and $[am_2^2, 2am_2]$
- 1) $2x + (m_1 + m_2)y - 2am_1m_2 = 0$ 2) $2x - (m_1 + m_2)y + 2am_1m_2 = 0$
3) $2x - (m_1 + m_2)y - 2am_1m_2 = 0$ 4) none
16. If the line joining the points $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ is parallel to $y = x$, then $t_1 + t_2 =$
- 1) $\frac{1}{2}$ 2) 4 3) $\frac{1}{4}$ 4) 2

MULTI CORRECT CHOICE TYPE:

17. Equation of a line passing through the point (2,3) and inclined at an angle of $\tan^{-1}\left(\frac{1}{2}\right)$ with the line $y + 2x = 5$ is

- 1) $y = 3$ 2) $x = 2$ 3) $3x + 4y - 18 = 0$ 4) $4x + 3y - 17 = 0$

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

18. The equation of the line intersecting the y - axis at a distance of 2 units above the origin and making an angle of 30° with the positive direction of the x - axis is

- 1) $x - \sqrt{3}y + 2 = 0$ 2) $x - \sqrt{3}y + 2\sqrt{3} = 0$
 3) $\sqrt{3}x - y + 2\sqrt{3} = 0$ 4) None of these

19. The equation of the line passing through the points $\left(ct_1, \frac{c}{t_1}\right), \left(ct_2, \frac{c}{t_2}\right)$ is

- 1) $x + t_1 t_2 y = c(t_1 + t_2)$ 2) $y + t_1 t_2 x = c(t_1 + t_2)$
 3) $x - t_1 t_2 y = c(t_1 + t_2)$ 4) $y - t_1 t_2 x = c(t_1 + t_2)$

MATRIX MATCH TYPE:**20. Column - I****Column - II**

- | | |
|---|----------------------|
| a) Line passing through (-4,3) and having intercepts in the ratio 5 : 3 | 1) $2x - 5y + 4 = 0$ |
| b) Line passing through P(2,-5) such that p bisects the part intercepted between the axes | 2) $x - 2y - 20 = 0$ |
| c) Line parallel to $2x - 3y + 5 = 0$ with x - intercept $2/5$ is | 3) $3x + 5y = 3$ |
| d) Line perpendicular to $5x + 2y + 7 = 0$ with y - intercept $4/5$ is | 4) $10x - 15y = 4$ |

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

21. The equation of the straight line joining the points $(c \cos \alpha, c \sin \alpha)$ and $(c \cos \beta, c \sin \beta)$ is

- 1) $x \cos \frac{(\alpha + \beta)}{2} + y \sin \frac{(\alpha + \beta)}{2} = c \cos \frac{(\alpha + \beta)}{2}$ 2) $x \cos \frac{(\alpha + \beta)}{2} + y \sin \frac{(\alpha + \beta)}{2} = c \cos \frac{(\alpha - \beta)}{2}$
 3) $x \cos \frac{(\alpha + \beta)}{2} + y \sin \frac{(\alpha - \beta)}{2} = c \cos \frac{(\alpha - \beta)}{2}$ 4) none

22. If $4a^2 + 9b^2 - c^2 + 12ab = 0$, then the set of lines $ax + by + c = 0$ passes through the point

- 1) (1, 2) (-1, -2) 2) (2, 3) (-2, -3) 3) (2, -3), (-2, 3) 4) (1, -2), (-1, 2)

23. If the slope of the line $\left(\frac{1}{a} + \frac{k}{b}\right)x + \left(\frac{1}{b} + \frac{k}{a}\right)y - (1+k) = 0$ is -1 then the value of k is]
 1) 2 2) -1 3) 1 4) -2
24. $P(\alpha, \beta)$ lies on the line $y = 6x - 1$ and $Q(\beta, \alpha)$ lies on the line $2x - 5y = 5$. Then the equation of the line PQ is
 1) $2x + y = 3$ 2) $3x + 2y = 5$ 3) $x + y = 6$ 4) $3x + y = 7$

Synopsis

Date: 23-08-2010

PERPENDICULAR FORM OR NORMAL FORM OF A LINE :

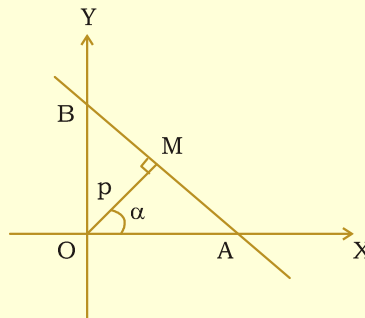
Definition 1 : The equation of the straight line which is at a distance of p units from the origin and $\alpha, 0 \leq \alpha < 360^\circ$, is the angle made by the perpendicular with positive direction of x -axis is $x \cos \alpha + y \sin \alpha = p$.

Proof : Let M be the projection of O on the line.

$$\therefore OM = p, \angle MOX = \alpha.$$

$$\therefore M = (p \cos \alpha, p \sin \alpha)$$

Case (i) : Suppose the given line is an inclined line.



$$\text{Slope of } \overline{OM} = \frac{p \sin \alpha}{p \cos \alpha} = \frac{\sin \alpha}{\cos \alpha}$$

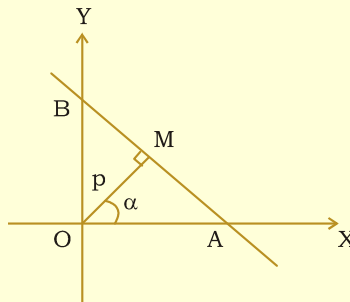
$$\text{Slope of the line} = -\frac{\cos \alpha}{\sin \alpha}$$

$$\text{Equation of the line is } y - p \sin \alpha = -\frac{\cos \alpha}{\sin \alpha}(x - p \cos \alpha)$$

$$\Rightarrow y \sin \alpha - p \sin^2 \alpha = -x \cos \alpha + p \cos^2 \alpha \Rightarrow x \cos \alpha + y \sin \alpha = p(\cos^2 \alpha + \sin^2 \alpha)$$

$$\Rightarrow x \cos \alpha + y \sin \alpha = p$$

Case (ii) : Suppose the given line is a horizontal line.



$$\therefore \alpha = 90^\circ \text{ or } 270^\circ$$

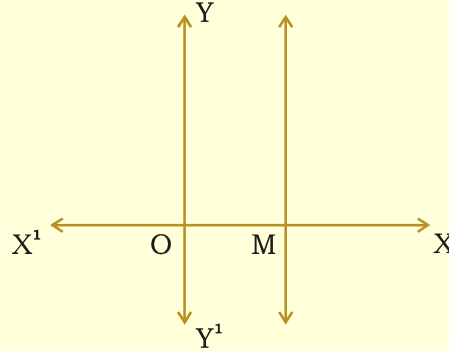
If $\alpha = 90^\circ$, then $M = (0, p)$

$\therefore$ Equation of the line is $y = p \Rightarrow x \cos 90^\circ + y \sin 90^\circ = p \Rightarrow x \cos \alpha + y \sin \alpha = p$

If $\alpha = 270^\circ$, then $M = (0, -p)$ Equation of the line is $y = -p \Rightarrow -y = p$

$\Rightarrow x \cos 270^\circ + y \sin 270^\circ = p \Rightarrow x \cos \alpha + y \sin \alpha = p$

Case (iii) : Suppose the given line is a vertical line.



$\therefore \alpha = 0^\circ$ or 180°

If $\alpha = 0^\circ$, then $M = (p, 0)$

Equation of the line is $x = p$

$\Rightarrow x \cos 0^\circ + y \sin 0^\circ = p \Rightarrow x \cos \alpha + y \sin \alpha = p$

If $\alpha = 180^\circ$, then $M = (-p, 0)$

Equation of the line is $x = -p$

$\Rightarrow -x = p \Rightarrow x \cos 180^\circ + y \sin 180^\circ = p$

$\Rightarrow x \cos \alpha + y \sin \alpha = p.$

$\therefore$ Equation of the required line is $x \cos \alpha + y \sin \alpha = p.$

Definition 2 : The equation of the line $ax + by + c = 0$ in perpendicular form is

$$\frac{a}{\sqrt{a^2 + b^2}}x + \frac{b}{\sqrt{a^2 + b^2}}y = \frac{-c}{\sqrt{a^2 + b^2}}, \text{ if } c \leq 0;$$

$$\frac{(-a)}{\sqrt{a^2 + b^2}}x + \frac{(-b)}{\sqrt{a^2 + b^2}}y = \frac{c}{\sqrt{a^2 + b^2}}, \text{ if } c > 0.$$

Case (i) : Suppose $c \leq 0$. Equation of the line is $ax + by + c = 0$

$\Rightarrow ax + by = -c$

$$\Rightarrow \frac{a}{\sqrt{a^2 + b^2}}x + \frac{b}{\sqrt{a^2 + b^2}}y = \frac{-c}{\sqrt{a^2 + b^2}}$$

$$\Rightarrow x \cos \alpha + y \sin \alpha = p \text{ where } \cos \alpha = \frac{a}{\sqrt{a^2 + b^2}},$$

$$\sin \alpha = \frac{b}{\sqrt{a^2 + b^2}} \text{ and } p = \frac{-c}{\sqrt{a^2 + b^2}} \geq 0$$

Case (ii) : Suppose $c > 0$.

Equation of the line is $ax + by + c = 0$

$$\Rightarrow -ax - by = c$$

$$\Rightarrow \frac{-a}{\sqrt{a^2 + b^2}}x + \frac{-b}{\sqrt{a^2 + b^2}}y = \frac{c}{\sqrt{a^2 + b^2}}$$

$$\Rightarrow x \cos \alpha + y \sin \alpha = p \text{ where } \cos \alpha$$

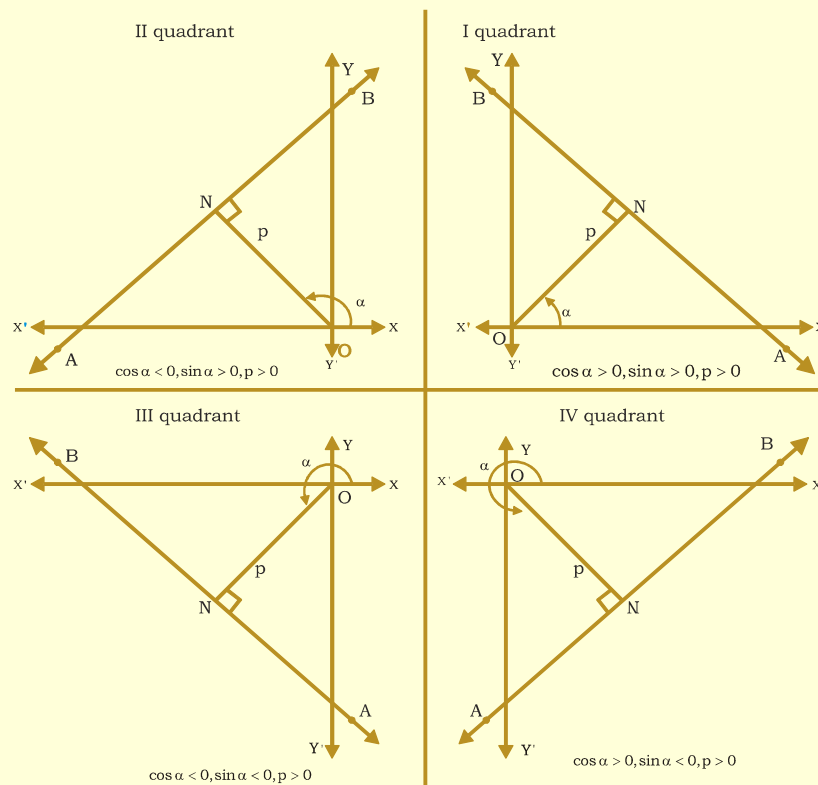
$$= \frac{-a}{\sqrt{a^2 + b^2}}, \sin \alpha = \frac{-b}{\sqrt{a^2 + b^2}} \text{ and } p = \frac{c}{\sqrt{a^2 + b^2}} > 0.$$

Note :- The perpendicular distance from origin to the line $ax + by + c = 0$ is $\frac{|c|}{\sqrt{a^2 + b^2}}$

Note :-

1. Here p is always taken as positive and α is measured from positive direction of x - axis in anticlockwise direction between 0 and 2π (i.e, $0 \leq \alpha < 2\pi$).
2. $(\text{coefficient of } x)^2 + (\text{coefficient of } y)^2 = \cos^2 \alpha + \sin^2 \alpha = 1$

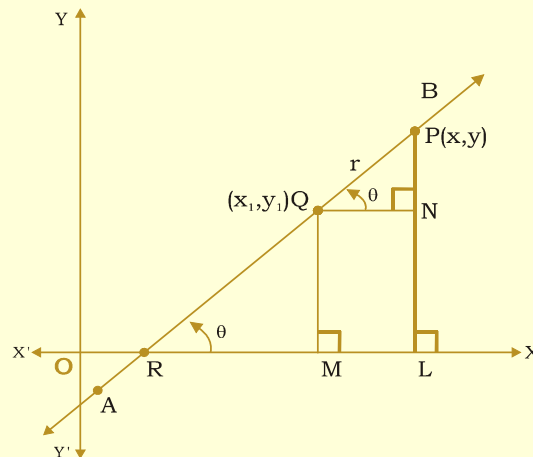
3.



SYMMETRIC FORM

Theorem : The equation of the straight line passing through (x_1, y_1) and making an angle θ with the positive direction of x - axis is $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$ where r is the directed distance between the points (x, y) and (x_1, y_1) .

Proof : Let AB be a line which passes through the point Q (x_1, y_1) and meet x-axis at R and makes an angle θ with the positive direction of x - axis.



Let P (x, y) be any point on the line at a distance r from Q.

Draw PL and QM are perpendiculars from P and Q on x - axis respectively and draw QN perpendicular on PL. Then $QN = ML = OL - OM = x - x_1$ and

$$PN = PL - NL = PL - QM = y - y_1$$

$$\text{from } \triangle PQN, \cos \theta = \frac{QN}{PQ} = \frac{x - x_1}{r} \text{ or } \frac{x - x_1}{\cos \theta} = r \quad \dots(1)$$

$$\text{and } \sin \theta = \frac{PN}{PQ} = \frac{y - y_1}{r} \text{ or } \frac{y - y_1}{\sin \theta} = r \quad \dots(2)$$

$$\text{From (1) and (2), we get } \frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$$

Corollary :

$$1. \because \frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r \text{ then } \begin{cases} x = x_1 + r \cos \theta \\ y = y_1 + r \sin \theta \end{cases}$$

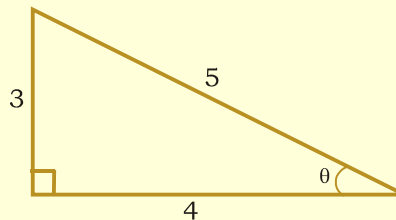
parametric equations of straight line AB.

2. If P point above Q (x_1, y_1) then r is positive then co-ordinates of P are $(x_1 + r \cos \theta, y_1 + r \sin \theta)$ and if P below Q (x_1, y_1) then r is negative then co-ordinates of P are $(x_1 - r \cos \theta, y_1 - r \sin \theta)$.

Illustration : The slope of a straight line through A(3,2) is $\frac{3}{4}$. Find the co-ordinates of the points on the line that are 5 units away from A.

Sol:- Let straight line makes an angle θ with positive direction of x - axis,

$$\text{then } \tan \theta = \frac{3}{4}$$



$$\therefore \sin \theta = \frac{3}{5} \text{ and } \cos \theta = \frac{4}{5}$$

$$\therefore \text{Equation of the straight line through A(3,2) in parametric form is } \frac{x-3}{\cos \theta} = \frac{y-2}{\sin \theta} = \pm 5$$

$$\therefore x = 3 \pm 5 \cos \theta = 3 \pm 5 \times \frac{4}{5} = 3 \pm 4 = 7 \text{ or } -1 \text{ and}$$

$$y = 2 \pm 5 \sin \theta = 2 \pm 5 \times \frac{3}{5} = 2 \pm 3 = 5 \text{ or } -1$$

Hence the co-ordinates of the points are (7,5) and (-1,-1).

Worksheet - 6

Date: 23-08-2010

LEVEL - 1

SINGLE CORRECT CHOICE TYPE:

- The distance of the straight line from the origin is 6 and angle made by the normal from the origin to the line with the positive direction of x - axis is 150° , then the equation of the straight line is
 - $\sqrt{3}x - y + 12 = 0$
 - $x - \sqrt{3}y + 12 = 0$
 - $\sqrt{3}y - x + 12 = 0$
 - $\sqrt{3}x - y - 12 = 0$
- The equation of the straight line whose distance from origin is 4 and normal from the origin to straight line makes an angle of 135° with the x - axis in positive direction is
 - $x + y = 4\sqrt{2}$
 - $x - y = 4\sqrt{2}$
 - $x - y = -4\sqrt{2}$
 - None of these
- The normal form of the line $4x - 3y + 12 = 0$ is
 - $\frac{-4}{5}x + \frac{3}{5}y = \frac{12}{5}$
 - $\frac{4}{5}x - \frac{3}{5}y = \frac{12}{5}$
 - $\frac{4}{5}x + \frac{3}{5}y = -\frac{12}{5}$
 - None of these

4. The equation of the straight line in the symmetric form having the given slope $\frac{1}{-\sqrt{3}}$ and passing through the point $(-2, 0)$ is

$$1) \frac{x+2}{-1/2} = \frac{y-0}{\sqrt{3}/2}$$

$$2) \frac{x-2}{1/2} = \frac{y-0}{1/\sqrt{3}}$$

$$3) \frac{x+2}{1/2} = \frac{0-y}{1/\sqrt{2}}$$

$$4) \frac{x+2}{-\sqrt{3}/2} = \frac{y-0}{1/2}$$

MULTI CORRECT CHOICE TYPE:

5. The equation of the line $ax + by + c = 0$ in the perpendicular form is

$$1) \frac{a}{\sqrt{a^2 + b^2}} \cdot x + \frac{b}{\sqrt{a^2 + b^2}} \cdot y = \frac{-c}{\sqrt{a^2 + b^2}} \text{ if } c \leq 0$$

$$2) \frac{-a}{\sqrt{a^2 + b^2}} \cdot x + \frac{(-b)}{\sqrt{a^2 + b^2}} \cdot y = \frac{c}{\sqrt{a^2 + b^2}} \text{ if } c > 0$$

$$3) \frac{a}{\sqrt{a^2 + b^2}} \cdot x + \frac{(-b)}{\sqrt{a^2 + b^2}} \cdot y = \frac{c}{\sqrt{a^2 + b^2}} \text{ if } c \leq 0$$

$$4) \frac{a}{\sqrt{a^2 + b^2}} \cdot x + \frac{b}{\sqrt{a^2 + b^2}} \cdot y = \frac{c}{\sqrt{a^2 + b^2}} \text{ if } c > 0$$

REASONING TYPE:

6. Statement I : Normal form of a straight line $4x - 3y + 12 = 0$ is $\frac{-4}{5}x + \frac{3}{5}y = \frac{12}{5}$

Statement II : Perpendicular distance from $(0,0)$ to $4x - 3y + 12 = 0$ is $\frac{12}{5}$

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

COMPREHENSION TYPE:

Perpendicular distance from (x_1, y_1) to $ax + by + c = 0$ is $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

7. The perpendicular distance from origin to the line $x - 3y + 4 = 0$ is

$$1) \frac{-4}{\sqrt{10}}$$

$$2) \frac{2}{\sqrt{10}}$$

$$3) \frac{-2}{\sqrt{10}}$$

$$4) \frac{4}{\sqrt{10}}$$

8. Transform the equation $\frac{x}{a} + \frac{y}{b} = 1$ is to normal form where $a > 0$, $b > 0$. If the perpendicular distance of the straight line from origin is p then $\frac{1}{p^2} =$
- 1) $\frac{1}{a} + \frac{1}{b}$ 2) $\frac{1}{a^2} + \frac{1}{b^2}$ 3) $\frac{b}{a^2} + \frac{a}{b^2}$ 4) $\frac{ab}{a+b}$
9. If p, q are the perpendicular from the origin to the line $x \sec \alpha + y \csc \alpha = a$ and $x \cos \alpha - y \sin \alpha = a \cos 2\alpha$, then $4p^2 + q^2 =$
- 1) a 2) $\frac{1}{a}$ 3) a^2 4) $\frac{1}{a^2}$

MATRIX MATCH TYPE:

10. Match the equations of the straight lines in the symmetric form in the following cases having the given slope and passing through the given point.

Column-I**Column-II**

- | | |
|---|---|
| a) Slope = $\sqrt{3}$, point = (2,3) | p) $\frac{x-2}{-1/\sqrt{2}} = \frac{y-3}{1/\sqrt{2}}$ |
| b) Slope = $-\frac{1}{\sqrt{3}}$, point = (+1,1) | q) $\frac{x-2}{1/2} = \frac{y-3}{\sqrt{3}/2}$ |
| c) Slope = -1, point = (1,1) | r) $\frac{x-1}{-1/2} = \frac{y-1}{\sqrt{3}/2}$ |
| d) Slope = -1, point = (2,3) | s) $\frac{x-1}{-1/\sqrt{2}} = \frac{y-1}{1/\sqrt{2}}$ |

INTEGER ANSWER TYRE:

11. The perpendicular distance from the origin to the line $3x + 4y - 15 = 0$ is _____

LEVEL - 2**SINGLE CORRECT CHOICE TYPE:**

12. The parametric equations of the line passing through (3, 2) and having inclination 135° are
- 1) $x = 3 + \frac{r}{\sqrt{2}}, y = 2 + \frac{r}{\sqrt{2}}$ 2) $x = 3 - \frac{r}{\sqrt{2}}, y = 2 + \frac{r}{\sqrt{2}}$
- 3) $x = 3 - \frac{r}{\sqrt{2}}, y = 2 - \frac{r}{\sqrt{2}}$ 4) none of these

13. The distance of the line $3x - y = 0$ from the point $(4, 1)$ measured along a line making an angle of 135° with the x-axis is

1) $11\sqrt{2}$ 2) $11/4$ 3) $11\sqrt{2}/4$ 4) none of these

MULTI CORRECT CHOICE TYPE:

14. The perpendicular form of the line $3x + 4y - 5 = 0$ is

1) $x \cos \alpha + y \sin \alpha = 1$ where $\cos \alpha = \frac{3}{5}$, $\sin \alpha = \frac{4}{5}$

2) $x \cos \alpha - y \sin \alpha = 1$ where $\cos \alpha = \frac{3}{5}$, $\sin \alpha = \frac{-4}{5}$

3) $x \sin \alpha + y \cos \alpha = 1$ where $\cos \alpha = \frac{3}{5}$, $\sin \alpha = \frac{4}{5}$

4) $x \cos \alpha + y \sin \alpha$ where $\cos \alpha = \frac{-3}{5}$, $\sin \alpha = \frac{4}{5}$

LEVEL - 3

SINGLE CORRECT CHOICE TYPE:

15. The equations of the lines on which the perpendiculars from the origin make 30° angle with x - axis and which form a triangle of area $\frac{50}{\sqrt{3}}$ with axes, are

1) $x + \sqrt{3}y \pm 10 = 0$ 2) $\sqrt{3}x + y \pm 10 = 0$ 3) $x \pm \sqrt{3}y - 10 = 0$ 4) None of these

16. The lines $x \cos \alpha + y \sin \alpha = p_1$ and $x \cos \beta + y \sin \beta = p_2$ will be perpendicular, if

1) $\alpha \pm \beta = \frac{\pi}{2}$ 2) $\alpha = \frac{\pi}{2}$ 3) $|\alpha - \beta| = \frac{\pi}{2}$ 4) $\alpha = \beta$

Reasoning Type:

17. Statement - I : The normal form of the line $x + y = \sqrt{2}$ is $x \cos \frac{\pi}{4} + y \sin \frac{\pi}{4} = 1$

Statement-II: Equation to normal form of a line is $x \cos \alpha + y \sin \alpha = p$ ($p > 0$)

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

LEVEL - 4

SINGLE CORRECT CHOICE TYPE:

18. A line is drawn through $P(3, 4)$ inclined at an angle 135° with x-axis. The points on the line on opposite sides of P at a distance $\sqrt{2}$ from it are

1) $(2, 5), (4, 3)$ 2) $(-2, -5), (-4, -3)$ 3) $(2, 5), (-4, -3)$ 4) none of these

19. The slope of a line through $A(1, 1)$ is 1. The point on the line at a distance $5\sqrt{2}$ from A is

1) $(-6, -6)$ 2) $(4, -4)$ 3) $(6, 6)$ 4) None of these

Reasoning Type:

20. *Statement - I* : If the slope of a line passing through A(2,3) is $\frac{3}{4}$ then the points on the line that are 5 units away from A are (6,6) & (-2,0)

Statement - II: If (x,y) is any point on the line and (r) is the distance from (x_1, y_1) to (x,y) then the parametric equations of the line are $x = x_1 + r \cos \theta, y = y_1 + r \sin \theta$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

21. The equation of a line whose perpendicular distance from the origin is 5 units and the angle between the positive direction of the x - axis and the perpendicular is 30° is
- 1) $\sqrt{3}x + y + 10 = 0$ 2) $\sqrt{3}x - y + 10 = 0$ 3) $\sqrt{3}x + y - 10 = 0$ 4) $-\sqrt{3}x + y + 10 = 0$
22. The equation of the line whose perpendicular distance from the origin is 3 units and the angle between the positive direction of x-axis and the perpendicular is 15° is
- 1) $(\sqrt{3} - 1)x + (\sqrt{3} + 1)y - 6\sqrt{2} = 0$ 2) $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y - 6\sqrt{2} = 0$
- 3) $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y + 6\sqrt{2} = 0$ 4) None of these
23. The equation of a line whose perpendicular distance from the origin is $\sqrt{8}$ units and the angle between the positive direction of the x-axis and the perpendicular is 135° is
- 1) $x + y + 4 = 0$ 2) $x - y + 4 = 0$ 3) $x + y - 4 = 0$ 4) None of these

Synopsis**Date: 24-08-2010****Illustration :**

The equation of the line whose perpendicular distance from the origin is 4 units and the angle which the normal makes with positive direction of x - axis is 15° .

- 1) $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$ 2) $(\sqrt{3} + 1)x - (\sqrt{3} - 1)y = 8\sqrt{2}$
- 3) $(\sqrt{3} - 1)x + (\sqrt{3} + 1)y = 8\sqrt{2}$ 4) $(\sqrt{3} - 1)x - (\sqrt{3} + 1)y = 8\sqrt{2}$

Solution :

Here, we are given $p = 4$ and $\alpha = 15^\circ$

$$\text{Now } \cos 15^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}} \text{ and } \sin 15^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

By the normal form above, the equation of the line is

$$x \cos 15^\circ + y \sin 15^\circ = 4 \text{ or } \frac{\sqrt{3}+1}{2\sqrt{2}}x + \frac{\sqrt{3}-1}{2\sqrt{2}}y = 4 \text{ or } (\sqrt{3}+1)x + (\sqrt{3}-1)y = 8\sqrt{2}$$

This is the required equation.

General Equation of a Line

$$Ax + By + C = 0$$

Different from of $Ax + By + C = 0$

I. Slope Intercept Form

$$\text{If } B \neq 0, \text{ then } Ax + By + C = 0 \quad \dots\dots(i)$$

can be written as

$$y = -\frac{A}{B}x - \frac{C}{B} \quad \dots\dots(ii)$$

$$\text{Compare (ii) with (iii) } y = mx + c \quad \dots\dots(iii)$$

$$\text{Slope of line } m = -\frac{A}{B}$$

$$\text{Intercept on y - axis } c = -\frac{C}{B}$$

II. Intercept form : If $C \neq 0$

$$\text{then } Ax + By + C = 0 \text{ can be written as } \dots\dots(i)$$

$$\frac{x}{\left(-\frac{C}{A}\right)} + \frac{y}{\left(-\frac{C}{B}\right)} = 1 \quad \dots\dots(ii)$$

Compare (ii) with (iii)

$$\frac{x}{a} + \frac{y}{b} = 1 \quad \dots\dots(iii)$$

$$\text{where } I_x = \text{Intercept on x - axis} = -\frac{C}{A}$$

$$I_y = \text{Intercept on y - axis} = -\frac{C}{B}$$

III. Normal Form

$$\text{Let } Ax + By + C = 0 \text{ can be written as } \dots\dots(i)$$

$$Ax + By = -C \quad \dots\dots(ii)$$

Compare (ii) with (iii)

$$x \cos \alpha + y \sin \alpha = p \quad \dots\dots(iii)$$

$$\text{so } \frac{A}{\cos \alpha} = \frac{B}{\sin \alpha} = \frac{-C}{p}$$

so $\cos \alpha = \frac{-PA}{C}$ and $\sin \alpha = \frac{-PB}{C}$. Using $\sin^2 \alpha + \cos^2 \alpha = 1$

$$\left(\frac{-PA}{C}\right)^2 + \left(\frac{-PB}{C}\right)^2 = 1 \Rightarrow p^2 = \frac{C^2}{A^2 + B^2} \Rightarrow p = \pm \frac{C}{\sqrt{A^2 + B^2}}$$

proper choice of signs is made so that p should be +ve

$$\text{so } \cos \alpha = \pm \frac{A}{\sqrt{A^2 + B^2}} / \sin \alpha = \pm \frac{B}{\sqrt{A^2 + B^2}}$$

POINT OF INTERSECTION OF TWO LINES

Let the equations of two lines be $a_1x + b_1y + c_1 = 0$ (i)

and $a_2x + b_2y + c_2 = 0$ (ii)

suppose these two lines intersect at a point $P(x_1, y_1)$. Then (x_1, y_1) satisfies each of the given equations.

$\therefore a_1x_1 + b_1y_1 + c_1 = 0$ and $a_2x_1 + b_2y_1 + c_2 = 0$ solving these two by cross -

multiplication, we get $\frac{x_1}{b_1c_2 - b_2c_1} = \frac{y_1}{c_1a_2 - c_2a_1} = \frac{1}{a_1b_2 - a_2b_1} \Rightarrow x_1 = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}$

$$y_1 = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$$

Hence, the coordinates of the point of intersection of (i) and (ii) are

$$\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right)$$

Example: Find the point of intersection of the lines $x - 3y + 6 = 0$, $2x + 3y - 10 = 0$.

Sol: Let $P(\alpha, \beta)$ be the point of intersection $\therefore \alpha - 3\beta + 6 = 0$, $2\alpha + 3\beta - 10 = 0$

$$\frac{\alpha}{30 - 18} = \frac{\beta}{12 + 10} = \frac{1}{3 + 6} \Rightarrow \frac{\alpha}{12} = \frac{\beta}{22} = \frac{1}{9} \Rightarrow \alpha = \frac{12}{9} = \frac{4}{3}, \beta = \frac{22}{9}$$

$\therefore$ Point of intersection $P = (4/3, 22/9)$.

Worksheet - 7

Date: 24-08-2010

LEVEL - 1

SINGLE CORRECT CHOICE TYPE:

- The slope of a straight line through $A(3, 2)$ is $3/4$ then the coordinates of the points in the line that are 25-units away from A are
 - $(23, -17)$ $(-17, -13)$
 - $(23, 17)$ $(-17, -13)$
 - $(-23, 17)$ $(-17, 13)$
 - $(23, -17)$ $(-17, 13)$
- A straight line drawn through $(1, 1)$ is such that it meets the line $x + y = 4$ at a distance of $\sqrt{2}$ units from that point. Then the inclination of that line is
 - π
 - $\pi/4$
 - $\pi/3$
 - $\pi/2$

3. The distance of the line $2x - 3y = 4$ from the point $(1, 1)$ in the direction of the line $x + y = 1$ is
- 1) $\sqrt{2}$ 2) $5\sqrt{2}$ 3) $1/\sqrt{2}$ 4) none of these
4. Reduce the equation $2x - 4y = 5$ into slope intercept form.

MULTI CORRECT CHOICE TYPE:

5. If p, q are lengths of perpendiculars from the origin to $x \cos \alpha - y \sin \alpha = \sin 2\alpha$; $x \sin \alpha + y \cos \alpha = \cos 2\alpha$ then $p^2 + q^2 =$
- 1) 2 2) 3 3) 4 4) 1

REASONING TYPE:

6. *Statement 1:* If the slope of a line passing through $A(2, 3)$ is $\frac{3}{4}$, then the points on the line that are 5 units away from A are $(6, 6)$ or $(-2, 0)$.
Statement 2: If (x, y) is any point on the line and $|r|$ is the distance from (x_1, y_1) to (x, y) then the parametric equations of the line are $x = x_1 + r \cos \theta$, $y = y_1 + r \sin \theta$
1. Both Statements are true, Statement II is the correct explanation of Statement I.
 2. Both Statements are true, Statement II is not correct explanation of Statement I.
 3. Statement I is true, Statement II is false.
 4. Statement I is false, Statement II is true.

COMPREHENSION TYPE:

Consider the equation $x + \sqrt{3}y + 4 = 0$. Then

7. The slope intercept form of the equation is
- 1) $y = \frac{1}{\sqrt{3}}x - \frac{4}{\sqrt{3}}$ 2) $y = \frac{-1}{\sqrt{3}}x + \frac{4}{\sqrt{3}}$ 3) $y = \frac{1}{\sqrt{3}}x + \frac{4}{\sqrt{3}}$ 4) $y = \frac{-1}{\sqrt{3}}x + \left(\frac{-4}{\sqrt{3}}\right)$
8. Reduce the given equation in to intercept form.
- 1) $\frac{x}{4} + \frac{y}{\sqrt{3}/4} = 1$ 2) $\frac{x}{-4} + \frac{y}{-4/\sqrt{3}} = 1$ 3) $\frac{x}{4/\sqrt{3}} + \frac{y}{4} = 1$ 4) $\frac{x}{4/\sqrt{3}} + \frac{y}{4/\sqrt{3}} = 1$
9. Reduce the given equation in to normal form
- 1) $\frac{1}{2}x - \frac{\sqrt{3}y}{2} = 2$ 2) $-\frac{1}{2}x - \frac{\sqrt{3}y}{2} = 2$ 3) $-\frac{1}{2}x + \frac{\sqrt{3}y}{2} = 2$ 4) $\frac{1}{2}x + \frac{\sqrt{3}y}{2} = 2$

MATRIX MATCH TYPE:**10. Column-I****Parallel lines**

a) $2x + 3y + 4 = 0, 2x + 3y - 5 = 0$

b) $x - 2y + 3 = 0, 4y - 2x + 1 = 0$

c) $x - y + 5 = 0, 3x - 3y - 5 = 0$

d) $x + y + 7 = 0, x + y + 8 = 0$

Column-II**Distance**

1) $\frac{7}{2\sqrt{5}}$

2) $\frac{1}{2\sqrt{2}}$

3) $\frac{9}{\sqrt{13}}$

4) $\frac{20}{3\sqrt{2}}$

5) $\frac{1}{\sqrt{2}}$

LEVEL - 2**SINGLE CORRECT CHOICE TYPE:**

11. A straight line drawn through the point (2, 1) is such that its point of intersection with the line $y - 2x + 6 = 0$ is at a distance $3\sqrt{2}$ from this point, then the inclination of the line is

1) $\pi/6$

2) $\pi/4$

3) $\pi/3$

4) $\pi/2$

12. If the straight line through the point P(3, 4) makes an angle $\frac{\pi}{6}$ with the x-axis and meets the line $3x + 5y + 1 = 0$ at Q, the length PQ is

1) $30(5 + 3\sqrt{3})$

2) $30(3\sqrt{3} - 5)$

3) $3(30\sqrt{3} - 5)$

4) none of these

LEVEL - 3**SINGLE CORRECT CHOICE TYPE:**

13. The equation of the straight line in the symmetric form having the given slope $\frac{1}{-\sqrt{3}}$ and passing through the point (1, 1) is

1) $\frac{x+2}{-1/2} = \frac{y-0}{\sqrt{3}/2}$

2) $\frac{x-2}{1/2} = \frac{y-0}{1/\sqrt{3}}$

3) $\frac{x+2}{1/2} = \frac{0-y}{1/\sqrt{2}}$

4) None of these

14. The parametric equations of the line passing through (3, 2) and having inclination 135° are

1) $x = 3 + \frac{r}{\sqrt{2}}, y = 2 + \frac{r}{\sqrt{2}}$

2) $x = 3 - \frac{r}{\sqrt{2}}, y = 2 + \frac{r}{\sqrt{2}}$

3) $x = 3 - \frac{r}{\sqrt{2}}, y = 2 - \frac{r}{\sqrt{2}}$

4) none

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

15. There are two parallel lines, one of which has the equation $3x + 4y = 2$. If the lines cut an intercept of length 5 on the line $x + y = 1$, then the equation of the other line is

1) $3x + 4y = \frac{\sqrt{6}-2}{2}$ 2) $3x + 4y = \frac{\sqrt{6}+2}{2}$ 3) $3x + 4y = 7$ 4) none of these

COMPREHENSION TYPE:

Consider the straight line $ax + by + c = 0$ then slope intercept form is

$$y = \left(-\frac{a}{b}\right)x + \left(\frac{-c}{b}\right) \text{ intercept form is } \frac{x}{(-c/a)} + \frac{y}{(-c/b)} = 1$$

$$\text{Normal form is } \frac{ax}{\sqrt{a^2 + b^2}} + \frac{by}{\sqrt{a^2 + b^2}} = \frac{-c}{\sqrt{a^2 + b^2}} \text{ (if } c < 0\text{)}$$

$$\left(\frac{-a}{\sqrt{a^2 + b^2}}\right)x + \left(-\frac{b}{\sqrt{a^2 + b^2}}\right)y = \frac{c}{\sqrt{a^2 + b^2}} \text{ (if } c > 0\text{)}$$

Based on the above information, answer the following questions.

16. The perpendicular form of the line $\sqrt{3}x + y - 4 = 0$ is

1) $x \cos \frac{5\pi}{6} + y \sin \frac{5\pi}{6} = 2$ 2) $x \cos \frac{5\pi}{6} + y \sin \frac{5\pi}{6} = 4$
 3) $x \cos \frac{\pi}{6} + y \sin \frac{\pi}{6} = 2$ 4) $x \cos \frac{11\pi}{6} + y \sin \frac{11\pi}{6} = 2$

17. The slope form of the line $3x - 4y + 6 = 0$ is

1) $y = \frac{3}{4}x + \frac{3}{2}$ 2) $y = \left(-\frac{3}{4}\right)x + \frac{3}{2}$ 3) $y = \left(-\frac{3}{4}\right)x - \left(\frac{3}{2}\right)$ 4) $y = \left(\frac{3}{4}\right)x - \frac{3}{2}$

18. The intercepts on x - and , y - axis respectively, of the line $4x + 3y - 9 = 0$ are

1) $\frac{-9}{4}, 3$ 2) $\frac{+9}{4}, 3$ 3) $\frac{-9}{4}, -3$ 4) $\frac{9}{4}, -3$

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

19. A straight line is drawn through the point (2, 1) is such that its point of intersection with the line $y - 2x + 6 = 0$ is at a distance $3\sqrt{2}$ from this point, then the inclination of the line is

1) $\pi/6$ 2) $\pi/4$ 3) $\pi/3$ 4) $\pi/2$

20. If the straight line through the point P(3, 4) makes an angle $\pi/6$ with the x-axis and meets the line $3x + 5y + 1 = 0$ at Q, the length PQ is

1) $30(5 - 3\sqrt{3})$ 2) $30(3\sqrt{3} - 5)$ 3) $3(30\sqrt{3} - 5)$ 4) none of these

Synopsis**Date: 25-08-2010****ANGLE BETWEEN TWO STRAIGHT LINES AND COLLINEAR POINTS****I. Angle between two lines :**

The acute angle between lines L_1 and L_2 with slopes m_1 and m_2 respectively is

given by $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$ as $m_1 m_2 \neq -1$.

Thus obtuse angle $\phi = 180^\circ - \theta$

Illustration :

If the angle between two lines is $\frac{\pi}{4}$ and slope of one of the lines is $\frac{1}{2}$, then the slope of the other line is

- 1) $3, -\frac{1}{3}$ 2) $2, -\frac{1}{2}$ 3) $1, -1$ 4) $\frac{1}{3}, -3$

Solution :

We know that the acute angle θ between two lines with slopes m_1 and m_2 is

given by $\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$ (i)

Let $m_1 = \frac{1}{2}, m_2 = m$ and $\theta = \frac{\pi}{4}$.

Now, putting these values in (i), we get $\tan \frac{\pi}{4} = \left| \frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} \right|$ or $1 = \left| \frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} \right|$, which

gives $\frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} = 1$ or $-\frac{m - \frac{1}{2}}{1 + \frac{1}{2}m} = -1$

Therefore $m = 3$ or $m = -\frac{1}{3}$

Hence, slope of the other line is 3 or $-\frac{1}{3}$.

Illustrations :

Line through the points (-2,6) and (4,8) is perpendicular to the line through (8,12) and (x,24) . Then the value of x is

1) 2

2) 3

3) 4

4) 5

Solution:

Slope of the line through the points (-2,6) and (4,8) is

$$m_1 = \frac{8-6}{4-(-2)} = \frac{2}{6} = \frac{1}{3}$$

Slope of the line through the points (8,12) and (x, 24) is

$$m_2 = \frac{24-12}{x-8} = \frac{12}{x-8}$$

Since two lines are perpendicular, $m_1 m_2 = -1$, which gives

$$\frac{1}{3} \times \frac{12}{x-8} = -1 \text{ or } x = 4$$

LINES PARALLEL AND PERPENDICULAR TO A GIVEN LINE**EQUATION OF A LINE PARALLEL TO A GIVEN LINE**

THEOREM : Prove that the equation of a line parallel to a given line $ax + by + c = 0$ is $ax + by + \lambda = 0$, where λ is a constant.

PROOF: Let m be the slope of the line $ax + by + c = 0$. Then, $m = -\frac{a}{b}$

Since the required line is parallel to the given line, the slope of the required line is also m.

Let c_1 be the y - intercept of the required line. Then its equation is

$$y = mx + c_1 \Rightarrow y = -\frac{a}{b}x + c_1 \Rightarrow c_1x + by - bc_1 = 0$$

$\Rightarrow ax + by + \lambda = 0$, where λ can be determined by some given condition.

EQUATION OF A LINE PERPENDICULAR TO A GIVEN LINE

THEOREM : Prove that the equation of a line perpendicular to a given line $ax + by + c = 0$ is $bx - ay + \lambda = 0$, where λ is a constant.

PROOF : Let m_1 be the slope of the given line and m_2 be the slope of a line perpendicular to the given line . Then,

$$m_1 = -\frac{a}{b} \text{ and } m_1 m_2 = -1 \Rightarrow m_2 = -\frac{1}{m_1} = \frac{b}{a}$$

Let c_2 be the y - intercept of the required line. Then its equation is

$$y = m_2x + c_2 \Rightarrow y = \frac{b}{a}x + c_2 \Rightarrow bx - ay + \lambda = 0, \text{ where } \lambda = ac_2 = \text{constant}$$

Illustration : Find the equation of the line which is parallel to $3x - 2y + 5 = 0$ and passes through the point $(5, -6)$.

Solution: The equation of any line parallel to the line $3x - 2y + 5 = 0$ is

$$3x - 2y + \lambda = 0 \quad \dots\dots(i)$$

This passes through $(5, -6)$

$$\therefore 3 \times 5 - 2 \times -6 + \lambda = 0 \Rightarrow \lambda = -27$$

putting $\lambda = -27$ in (i), we get.

$$3x - 2y - 27 = 0, \text{ which is the required equation.}$$

ALITER . The slope of the given line is $3/2$. Therefore, the slope of the required line is also $3/2$. Since the required line passes through $(5, -6)$, so its equation is

$$y + 6 = \frac{3}{2}(x - 5) \Rightarrow 3x - 2y - 27 = 0$$

Illustration: Find the equation of the straight line that passes through the point $(3, 4)$ and perpendicular to the line $3x + 2y + 5 = 0$.

Solution: The equation of a line perpendicular to $3x + 2y + 5 = 0$ is

$$2x - 3y + \lambda = 0$$

This passes through the point $(3, 4)$

$$\therefore 3 \times 2 - 3 \times 4 + \lambda = 0 \Rightarrow \lambda = 6$$

putting $\lambda = 6$ in (i), we get

$$2x - 3y + 6 = 0, \text{ which is the required equation.}$$

Worksheet - 8

Date: 25-08-2010

LEVEL - 1

SINGLE CORRECT CHOICE TYPE:

- The value of k if the lines $2x + ky - 10 = 0$ and $5x + 2y - 7 = 0$ are parallel is
 1) $\frac{4}{5}$ 2) $\frac{3}{5}$ 3) $\frac{5}{6}$ 4) $\frac{5}{2}$
- Equation of a straight line through the point $(2, 3)$ and perpendicular to straight line $4x + 7y = 9$ is
 1) $7x + 4y = 26$ 2) $7x - 4y + 2 = 0$ 3) $7x - 4y = 2$ 4) none of these
- The equation of the line through the point of intersection of the lines $x - 3y + 1 = 0$, $2x + 5y - 9 = 0$ and is parallel to the line $y = x + 2$ is
 1) $y = x + 3$ 2) $x - y = 5$ 3) $x - y = 1$ 4) none of these
- The acute angle between the lines $y = 4 + x$; $y = 7$ is
 1) 45° 2) 30° 3) 90° 4) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

MULTI CORRECT CHOICE TYPE:

5. If the acute angle between the lines $4x - y + 7 = 0$, $kx + 5y - 9 = 0$ is 45° . Then the value of K is

- 1) -3 2) -4 3) $\frac{25}{3}$ 4) $\frac{2}{3}$

REASONING TYPE:

6. *Statement -I:* If the acute angle between the lines $3x - y + 1 = 0$, $2x + ky + 5 = 0$ is 45° then the value of K is 1 or -4

Statement -II: If θ is the angle between the lines $a_1x + b_1y + c_1 = 0$,

$$a_2x + b_2y + c_2 = 0 \text{ then } \cos \theta = \left| \frac{a_1a_2 + b_1b_2}{\sqrt{a_1^2 + b_1^2} \sqrt{a_2^2 + b_2^2}} \right|$$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

COMPREHENSION TYPE:

The lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are parallel then $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ and the

lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are perpendicular then $a_1a_2 + b_1b_2 = 0$

7. If $2x + 3y + 5 = 0$, $kx + 6y + 7 = 0$ are parallel, then the value of K is

- 1) 2 2) 4 3) 6 4) -9

8. If $3x - ky - 2 = 0$, $2x + y + 2 = 0$ are perpendicular, then K =

- 1) $\frac{4}{3}$ 2) $\frac{4}{5}$ 3) $\frac{5}{3}$ 4) 6

9. If $5x + 3y + 7 = 0$, $kx - 7y + 8 = 0$ are perpendicular then K =

- 1) $\frac{21}{5}$ 2) $\frac{1}{3}$ 3) $\frac{5}{3}$ 4) -2

MATRIX MATCH TYPE:10. **Column-I****Column-II**

- a) If θ is the angle between two lines then another angle between the lines is

1) $\frac{3\pi}{4}$

- b) The angle between the lines $3x - y + 5 = 0$, $x + 3y - 2 = 0$ is

2) $(\pi - \theta)$

- c) If θ is the angle between $\frac{x}{a} + \frac{y}{b} = 1$, $\frac{x}{b} + \frac{y}{a} = 1$, then $\cos \theta =$

3) $\frac{\pi}{4}$

- d) Acute angle between the lines $ax + by = a + b$

4) $\frac{\pi}{2}$

$a(x - y) + b(x + y) = 2b$ is

5) $\frac{2ab}{a^2 + b^2}$

LEVEL - 2**SINGLE CORRECT CHOICE TYPE:**

11. If the angle between the lines $kx + y + 9 = 0$, $y - 3x = 4$ is 45° the value of k is
 1) $-1/2$ 2) $1/3$ 3) $1/2$ 4) $-1/3$
12. If θ is the angle between the line $\frac{x}{a} + \frac{y}{b} = 1$ and $\frac{x}{b} + \frac{y}{a} = 1$, then $\cos\theta$ is equal to
 1) $\frac{a^2 + b^2}{2ab}$ 2) $\frac{ab}{a^2 + b^2}$ 3) $\frac{2ab}{a^2 + b^2}$ 4) None

REASONING TYPE:

13. *Statement -I:* If the angle between the line $kx - y + 6 = 0$, $3x + 5y + 7 = 0$ is $\pi/4$ one value of k is 4

Statement -II: If θ is angle between the lines with slopes m_1 , m_2 then

$$\tan\theta = \frac{|m_1 - m_2|}{|1 + m_1 m_2|}.$$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

LEVEL - 3**SINGLE CORRECT CHOICE TYPE:**

14. If $x + 3y = 16$ is the perpendicular bisector of the line joining A, B and $B = (3, 1)$. Then A is
 1) $(5, -7)$ 2) $(5, 7)$ 3) $(4, 4)$ 4) $(4, -4)$
15. The equation of perpendicular bisectors of the sides AB and BC of the triangle ABC are $x - y - 5 = 0$ and $x + 2y = 0$ respectively and the point A is $(1, -2)$, then coordinates of the point C are
 1) $(1, 0)$ 2) $(0, 1)$ 3) $(5, 0)$ 4) $(0, 0)$
16. The angle between lines $y = (2 - \sqrt{3}x + 5)$ and $y = (2 + \sqrt{3})x - 7$ is
 1) 30° 2) 45° 3) 60° 4) 90°
17. A line perpendicular to the line $3x - 2y = 5$ cuts off an intercept 3 on the positive side of the x - axis, then
 1) The slope of the line is $2/3$ 2) The intercept on the y - axis is 2
 3) The intercept on the y - axis is 3 4) None

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

18. The equation of the line passing through the intersection of the line passing through the intersection of $x - \sqrt{3}y + \sqrt{3} - 1 = 0$ and $x + y - 2 = 0$ and making an angle of 15° with the first line is
 1) $x - y = 0$ 2) $x - y + 1 = 0$ 3) $y = 1$ 4) $\sqrt{3}x - y + 1 - \sqrt{3} = 0$

REASONING TYPE:

19. **Statement-1** : The two lines given by $ax + by = a + b$ and $a(x - y) + b(x + y) = 2b$ are inclined at $\frac{\pi}{4}$

Statement-2 : If θ is the acute angle between the lines $a_1x + b_1y + c_1 = 0$,

$$a_2x + b_2y + c_2 = 0, \text{ then } \tan\theta = \left| \frac{a_1b_2 - a_2b_1}{a_1a_2 + b_1b_2} \right|$$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

20. The image of the point A(1,2) by the line mirror $y = x$ is the point B and the image of B by the line mirror $y = 0$ is the point (α, β) . Find α and β .
21. A vertex of an equilateral triangle is (2, 3) and the opposite side is $x + y = 2$. Find the equation of other side.

Synopsis**Date: 26-08-2010****LENGTH OF PERPENDICULAR AND REFLECTION**

Length of perpendicular from (x_1, y_1) to straight line $ax + by + c = 0$ is given by

$$\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

If two lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ are parallel then $d = \left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$.

Illustrations :

The distance of the point (3,-5) from the line $3x - 4y - 26 = 0$ is

- 1) $\frac{1}{5}$ 2) $\frac{2}{5}$ 3) $\frac{3}{5}$ 4) $\frac{4}{5}$

Solution:

Given line is $3x - 4y - 26 = 0$

.....(i)

Comparing (1) with general equation of line $Ax + By + C = 0$, we get

$A = 3$, $B = -4$ and $C = -26$.

Given point is $(x_1, y_1) = (3, -5)$. The distance of the given point from given line is

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}} = \frac{|3 \cdot 3 + (-4)(-5) - 26|}{\sqrt{3^2 + (-4)^2}} = \frac{3}{5}$$

Illustrations :

The distance between the parallel lines $3x - 4y + 7 = 0$ and $3x - 4y + 5 = 0$ is

1) $\frac{1}{5}$

2) $\frac{2}{5}$

3) $\frac{3}{5}$

4) $\frac{4}{5}$

Solution :

Here $A = 3$, $B = -4$, $C_1 = 7$ and $C_2 = 5$. Therefore the required distance is

$$d = \frac{17 - 51}{\sqrt{3^2 + (-4)^2}} = \frac{2}{5}$$

Theorem : The ratio in which the line $ax + by + c = 0$, ($ab \neq 0$) divides the line segment

$$\overline{AB} \text{ joining the points } A(x_1, y_1) \text{ and } B(x_2, y_2) \text{ is } -\left(\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c}\right) = \frac{-L_{11}}{L_{22}}$$

Proof : Let the line $ax + by + c = 0$ divide $\overline{AB}$ in the ratio $m : n$ at P .

Then the coordinates of the point P are $\left[\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right]$

$$P \text{ lies on } ax + by + c = 0 \Leftrightarrow a\left[\frac{mx_2 + nx_1}{m+n}\right] + b\left[\frac{my_2 + ny_1}{m+n}\right] + c = 0$$

$$\Leftrightarrow a(mx_2 + nx_1) + b(my_2 + ny_1) + c(m+n) = 0 \Leftrightarrow m(ax_2 + by_2 + c) + n(ax_1 + by_1 + c) = 0$$

$$\therefore \frac{m}{n} = -\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} = \frac{-L_{11}}{L_{22}}$$

Note : The points $A(x_1, y_1)$, $B(x_2, y_2)$ lie on the

(a) same side of the line $ax + by + c = 0$, if $\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} > 0$, i.e. $\frac{L_{11}}{L_{22}} > 0$

(b) opposite sides of the line $ax + by + c = 0$ if $\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} < 0$ i.e. $\frac{L_{11}}{L_{22}} < 0$

Worksheet - 9**Date: 26-08-2010****LEVEL - 1****SINGLE CORRECT CHOICE TYPE:**

- The two points $(2, 1)$ and $(3, -1)$ with respect to the line $3x - 5y + 7 = 0$ lie
 - On the line
 - On the same side of line
 - On opposite sides of the line
 - none of these
- If the straight line $3x + 4y = 8$ is denoted by L , then the points $(2, -5)$ and $(-5, 2)$
 - Lie on the line L
 - Lie on the same side of L
 - Are equidistant from L
 - Lie on opposite sides of L

3. Let 'O' be the origin and $A = (3, -2)$, $B = (1, 2)$ and $C = (1, -2)$. Then the pair of points which are on the same side of the line $2x + 3y = 5$ are
 1) B, C 2) A, C 3) A, B 4) none of these
4. Equation of the line perpendicular to $4x + 7y + 9 = 0$ and such the triangle formed by it with the coordinate axes forms an area of 3.5 sq.units is
 1) $7x - 4y = 0$ 2) $4x - 7y = 0$ 3) $7x - 4y \pm 14 = 0$ 4) none of these

MULTI CORRECT CHOICE TYPE:

5. If p be the measure of the perpendicular segment from the origin on the line whose intercepts on the axes are 'a' and 'b', then which of the following is/are correct?

1) $\frac{1}{a^2} = \frac{1}{p^2} + \frac{1}{b^2}$ 2) $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$ 3) $p^2 (a^2 + b^2) = a^2 b^2$ 4) $p^2 (a^2 b^2) = a^2 + b^2$

REASONING TYPE:

6. *Statement I* : If p is perpendicular distance from the origin to the straight line

$$\frac{x}{a} + \frac{y}{b} = 1 \text{ then } \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{4p^2}.$$

Statement II : The perpendicular distance from origin to the line $ax + by + c = 0$ is

$$\frac{|c|}{\sqrt{a^2 + b^2}}$$

- Both Statements are true, Statement II is the correct explanation of Statement I.
 - Both Statements are true, Statement II is not correct explanation of Statement I.
 - Statement I is true, Statement II is false.
 - Statement I is false, Statement II is true.
7. *Statement I* : The reason in which the perpendicular through (4, 1) divides the line joining (2, -1), (6, 5) is 5:8.
Statement II : The ratio in which the line $ax + by + c = 0$ divides the line segment joining (x_1, y_1) , (x_2, y_2) is $(ax_1 + by_1 + c) : -(ax_2 + by_2 + c)$.
- Both Statements are true, Statement II is the correct explanation of Statement I.
 - Both Statements are true, Statement II is not correct explanation of Statement I.
 - Statement I is true, Statement II is false.
 - Statement I is false, Statement II is true.

COMPREHENSION TYPE:

The ratio in which the line $L \equiv ax + by + c = 0$ divided the line segment joining $A(x_1, y_1)$, $B(x_2, y_2)$ is $-L_{11} : L_{22}$ where $L_{11} = L(x_1, y_1) = ax_1 + by_1 + c$,
 $L_{22} = L(x_2, y_2) = ax_2 + by_2 + c$.

The point A, B lie in the same side or opposite sides of the line $L = 0$ according as L_{11} , L_{22} have the same sign or opposite signs.

Let $L \equiv ax + by + c = 0$ be a line and $A(x_1, y_1)$ be a point. Then A lies in the origin side of $L = 0$ if $cL_{11} > 0$ and A lies in the opposite to the origin side of $L = 0$ if $cL_{11} < 0$

Based on the above passage answer the following questions.

8. The ratio in which the line $3x - 4y + 5 = 0$ divides the line segment joining the points $(2, -4)$, $(-3, 1)$ is
 1) $26 : 9$ 2) $27 : 8$ 3) $24 : 7$ 4) $22 : 6$
9. The ratio in which the line joining the points $A(-1, -1)$ and $B(2, 1)$ divides the line joining $C(3, 4)$ and $D(1, 2)$ is
 1) $7 : 5$ internally 2) $7 : 5$ externally 3) $7 : 11$ internally 4) $7 : 11$ externally
10. The ratio in which the line joining the points $(2, 3)$ and $(4, -1)$ divides the line segment joining $(1, -2)$, $(-3, 2)$ is
 1) $7 : 5$ internally 2) $7 : 5$ externally 3) $7 : 11$ internally 4) $7 : 11$ externally

MATRIX MATCH TYPE:

11. The ratio in which line $x + 2y + 4 = 0$ divides the line segment joining points

Column-I

- a) $A(0, 0)$ and $B(2, 1)$
- b) $A(-2, 1)$ and $B(2, -1)$
- c) $A(-p, q)$ and $B(4q-3, p-q)$
- 4) $A(m, 2n)$ and $B(m-2, n + 1)$

Column-II

- 1) $1 : 4$
- 2) $1 : 1$
- 3) $1 : 2$
- 4) Points are at equidistant from the line.
- 5) $2 : 2$

LEVEL - 2

SINGLE CORRECT CHOICE TYPE:

12. Line L has intercepts a and b on the co-ordinate axes. When the axes are rotated through a given angle keeping the origin fixed, the same line has intercepts p and q then

- 1) $a^2 + b^2 = p^2 + q^2$
- 2) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$
- 3) $a^2 + p^2 = b^2 + q^2$
- 4) $\frac{1}{a^2} + \frac{1}{p^2} = \frac{1}{b^2} + \frac{1}{q^2}$

MULTI CORRECT CHOICE TYPE:

13. $L_1 : x + y - 3 = 0$, $L_2 : 2x - y = 0$ be two lines and $P_1(0,0)$ and $P_2(-2,1)$ be two points then
 - 1) P_1, P_2 are contained in the same angle formed by L_1 and L_2
 - 2) P_1, P_2 are contained in the adjacent angle formed by L_1 and L_2
 - 3) P_1, P_2 are contained in the opposite angle formed by L_1 and L_2
 - 4) The segment P_1, P_2 does not cut L_1 and L_2

LEVEL - 3**SINGLE CORRECT CHOICE TYPE:**

14. If the algebraic sum of the perpendicular distances from the points (2, 0), (0, 2) and (1, 1) to a variable straight line be zero, then the line passes through the point
 1) (-1, 1) 2) (1, 1) 3) (1, -1) 4) (-1, -1)
15. The points (-1, 1) and (1, -1) are symmetrical about the line
 1) $y + x = 0$ 2) $y = x$ 3) $x + y = 1$ 4) none of these

REASONING TYPE:

16. *Statement I:* If a transversal cuts the side $\overline{BC}$, $\overline{CA}$ and $\overline{AB}$ of a triangle in D, E, F respectively then $\frac{BD}{DC} \times \frac{CE}{EA} \times \frac{AF}{FB} = -1$.
- Statement II:* The ratio in which line L, $ax + by + c = 0$ divides the line segment joining $A(x_1, y_1)$, $B(x_2, y_2)$ is $\frac{-L_{11}}{L_{22}}$.
- Both Statements are true, Statement II is the correct explanation of Statement I.
 - Both Statements are true, Statement II is not correct explanation of Statement I.
 - Statement I is true, Statement II is false.
 - Statement I is false, Statement II is true.

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

17. The ratio in which the straight line $5x - 6y - 21 = 0$ divides the line segment joining the points (4, -1) and (2, 1) is
 1) 17 : 5 2) 5 : 17 3) -5 : 17 4) -17 : 5
18. The ratio in which the line joining the points A(1, -1) and B(2, 1) divides the line joining C(3, 4) and D(1, 2) is
 1) 7 : 5 2) -7 : 5 3) 5 : 7 4) 5 : -7

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

19. The ratio in which the line $3x + 4y = 8$ divides the line joining the points (0, 0) and (1, 1) is
 1) 8 : 1 internally 2) 8 : 1 externally 3) 1 : 8 internally 4) 1 : 8 externally
20. Let $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ be the vertices of the triangle ABC such that $x_3 = lx_1 + mx_2$, $y_3 = ly_1 + my_2$ where $l + m < 0$ then, origin
 1) Lies inside ΔABC 2) lies outside ΔABC
 3) and C lies on the same side of AB 4) Origin may lie inside ΔABC

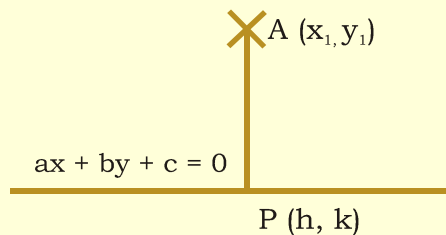
Synopsis

Date: 27-08-2010

1. Foot of the perpendicular :

Statement : If (h, k) is the foot of the perpendicular from (x_1, y_1) to the line

$$ax + by + c = 0 \quad (a \neq 0, b \neq 0), \text{ then } \frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{-(ax_1 + by_1 + c)}{a^2 + b^2}$$

Proof : Let $A(x_1, y_1)$, $P(h, k)$.

$$P \text{ lies in } ax + by + c = 0 \Rightarrow ah + bk + c = 0 \Rightarrow ah + bk = -c$$

$$\text{Slope of } \overline{AP} \text{ is } \frac{k - y_1}{h - x_1}. \text{ Slope of given line is } -\frac{a}{b}$$

$$\overline{AP} \text{ is perpendicular to the given line} \Rightarrow \left(\frac{k - y_1}{h - x_1} \right) \left(-\frac{a}{b} \right) = -1 \Rightarrow \frac{k - y_1}{b} = \frac{h - x_1}{a}$$

By the law of multipliers in ratio and proportion,

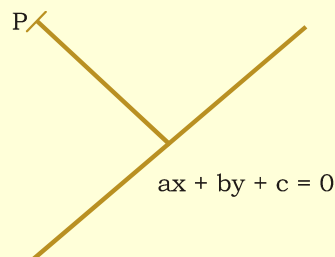
$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{a(h - x_1)}{a^2} = \frac{b(k - y_1)}{b^2} = \frac{a(h - x_1) + b(k - y_1)}{a^2 + b^2}$$

$$\left[\because \frac{x}{y} = \frac{z}{w} = \frac{x+z}{y+w} \right] = \frac{ah + bk - ax_1 - by_1}{a^2 + b^2} = \frac{-ax_1 - by_1 - c}{a^2 + b^2} = \frac{-(ax_1 + by_1 + c)}{a^2 + b^2}$$

2. PERPENDICULAR DISTANCE (LENGTH OF THE PERPENDICULAR) :

The perpendicular distance from a point $P(x_1, y_1)$ to the line $ax + by + c = 0$ is

$$\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Proof : Let the axes be translated to the point $P(x_1, y_1)$.Let (X, Y) be the new coordinates of (x, y) . Then $x = X + x_1$, $y = Y + y_1$

$\therefore$ The transformed equation of the given line is $a(X + x_1) + b(Y + y_1) + c = 0$
 $\Rightarrow aX + bY + (ax_1 + by_1 + c) = 0$.

$\therefore$ The perpendicular distance from the new origin P to the line is $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$.

1. IMAGE OF A POINT :

Definition :

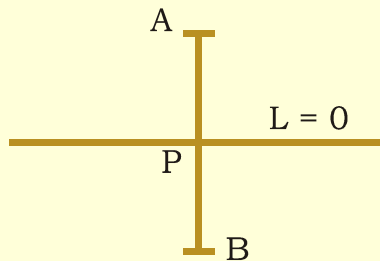
Let A, P be two points. A point B ($\neq A$) on the line $\overline{AP}$ such that $AP = PB$ is called image of A with respect to the point P.



Note : The image of (x_1, y_1) with respect to the origin is $(-x_1, -y_1)$

Definition :

Let A be a point which is not in the line $L = 0$ and P be the foot of perpendicular from A to the line $L = 0$. A point B ($\neq A$) in the line $\overline{AP}$ such that $AP = PB$ is called image of A with respect to the line $L = 0$.



Note 1 : The image of (x_1, y_1) with respect to the x-axis is $(x_1, -y_1)$

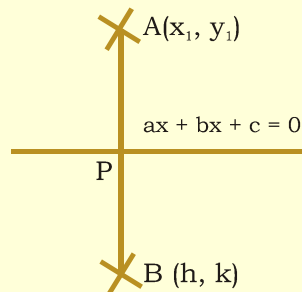
Note 2 : The image of (x_1, y_1) with respect to the y-axis is $(-x_1, y_1)$

Note 3 : The image of (x_1, y_1) with respect to the line $y = x$ is (y_1, x_1)

2. Theorem :- If (h, k) is the image of (x_1, y_1) with respect to the line $ax + by + c = 0$

$(a \neq 0, b \neq 0)$, then $\frac{h - x_1}{a} = \frac{k - y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}$

Proof : Let $A(x_1, y_1)$, $B(h, k)$.



Midpoint of $\overline{AB}$ is $P = \left(\frac{x_1 + h}{2}, \frac{y_1 + k}{2} \right)$

B is the image of A $\Rightarrow$ midpoint P lies on $ax + by + c = 0$

$$\Rightarrow a\left(\frac{x_1+h}{2}\right) + b\left(\frac{y_1+k}{2}\right) + c = 0 \Rightarrow ax_1 + by_1 + ah + bk + 2c = 0$$

$$\Rightarrow ah + bk = -ax_1 - by_1 - 2c$$

Slope of $\overline{AB}$ is $\frac{k-y_1}{h-x_1}$. Slope of the given line is $-\frac{a}{b}$.

$$\overline{AB} \text{ is perpendicular to the given line} \Rightarrow \left(\frac{k-y_1}{h-x_1}\right)\left(-\frac{a}{b}\right) = -1 \Rightarrow \frac{h-x_1}{a} = \frac{k-y_1}{b}$$

By the law of multipliers in ratio and proportion,

$$\begin{aligned} \frac{h-x_1}{a} = \frac{k-y_1}{b} &= \frac{a(h-x_1) + b(k-y_1)}{a^2 + b^2} = \frac{ah + bk - ax_1 - by_1}{a^2 + b^2} \\ &= \frac{-ax_1 - by_1 - 2c - ax_1 - by_1}{a^2 + b^2} \quad (ah + bk = -ax_1 - by_1 - 2c) = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2} \end{aligned}$$

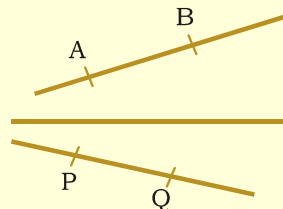
Note 1 : The image of (x_1, y_1) with respect to the line $x = y$ is (y_1, x_1)

Note 2 : The image of (x_1, y_1) with respect to the line $x + y = 0$ is $(-y_1, -x_1)$

3. IMAGE LINE :

Definition :

Let $L_1 = 0$, $L = 0$ be two lines and A, B be two points on $L_1 = 0$. If P, Q are the images of A, B with respect to $L = 0$ respectively, then the line $\overline{PQ}$ is called image line of $L_1 = 0$ with respect to $L = 0$.



Worksheet - 10

Date: 27-08-2010

LEVEL - 1

SINGLE CORRECT CHOICE TYPE:

- The image of the line $x = 1$ in y -axis is
 - $x = -1$
 - $x - 1 = 0$
 - $y = -1$
 - $y - 1 = 0$
- The foot of the perpendicular from $(2,3)$ to the line $4x - 5y + 8 = 0$ is
 - $\left(-\frac{78}{41}, \frac{-128}{41}\right)$
 - $\left(\frac{128}{41}, \frac{78}{41}\right)$
 - $\left(\frac{-128}{41}, \frac{-78}{41}\right)$
 - $\left(\frac{78}{41}, \frac{128}{41}\right)$

3. The foot of the perpendicular from $(-3, 7)$ to the line $3x - 4y + 12 = 0$ is
 1) $(0, 3)$ 2) $(3, 0)$ 3) $(-3, 0)$ 4) $(0, -3)$
4. The foot of the perpendicular from the origin on $3x + 4y - 25 = 0$ is
 1) $(3, 4)$ 2) $(8, 6)$ 3) $(-8, 6)$ 4) $(6, -8)$

MULTI CORRECT CHOICE TYPE:

5. Let $P(3, 4)$ be a point on the plane consider a line $L \equiv 3x + 4y + 25 = 0$ is a line on the same plane. then
 1) Foot of the perpendicular from 'P' on to the line L is $(-3, -4)$
 2) Image of P w.r.t $L = 0$ is $(-9, -12)$
 3) Midpoint of P and image of P w.r.t $L = 0$ is $(-3, -4)$
 4) Equation of $\overline{PQ}$ is $4x + 3y = 0$

REASONING TYPE:

6. Statement I : Image of the point $(6, 1)$ in the line $x + y = 0$ is $(-1, -6)$

Statement II : Image of a point $p(\alpha, \beta)$ in the line $ax + by + c = 0$ is $Q(\alpha^1, \beta^1)$, if

$$\left(\frac{\alpha + \alpha^1}{2}, \frac{\beta + \beta^1}{2} \right) \text{ lies on the given line}$$

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

COMPREHENSION TYPE:

The image of (a, b) with respect to $x = y$ is (b, a) .

The image of (a, b) w.r.t. $x + y = 0$ is $(-b, -a)$

The image of (a, b) w.r.t. $(0, 0)$ is $(-a, -b)$

The image of (x_1, y_1) w.r.to (x, y) is $(2x - x_1, 2y - y_1)$

The reflection of $ax + by + c = 0$ in the line $lx + my + n = 0$ is

$$(x + y + c)(l^2 + m^2) - 2(l + m)(lx + my + n) = 0.$$

Based on the above information, answer the following questions.

7. The image of $P(2, -3)$ with respect to $x = y$ is $Q(x, y)$ and image of $Q(x, y)$ w.r.t $(0, 0)$ is $R(x, y)$. Then co-ordinates of R are
 1) $(-3, 2)$ 2) $(3, 2)$ 3) $(3, -2)$ 4) $(2, 3)$
8. The image of $A(4, 6)$ w.r.to $(3, 2)$ is
 1) $(2, 2)$ 2) $(2, -2)$ 3) $(-2, 2)$ 4) $(-2, -2)$
9. The images of $2x + y - 1 = 0$ with respect to $x + y + 1 = 0$ is
 1) $7x + y + 1 = 0$ 2) $x + 7y - 11 = 0$ 3) $7x + y - 11 = 0$ 4) $7x + y + 7 = 0$

MATRIX MATCH TYPE:**10 Column-I**

- a) The reflection of (2,3) w.r.t. (5,3) is
 b) The foot of the perpendicular from (0,0) on $3x + 4y - 25 = 0$ is
 c) The points (-1,1) and (1,-1) are symmetrical about the line
 d) The line $x + y = 4$ is the perpendicular bisector of $\overline{AB}$ where $A(3, -3)$, then 'B' is

Column-II

- 1) (7,1)
 2) (8,3)
 3) y - axis
 4) $x = y$

LEVEL - 2**SINGLE CORRECT CHOICE TYPE:**

11. If the point (1, 2) is reflected through the origin and then through the line $x = y$, then the new coordinates of the point are
 1) (-2, -1) 2) (2, 1) 3) (-1, 2) 4) (1, 2)
 12. If a ray travelling along the line $x = 1$ gets reflected from the line $x + y = 1$, then the equation of the line along which the reflected ray travels is
 1) $y = 0$ 2) $x - y = 1$ 3) $x = 0$ 4) none of these

REASONING TYPE:

13. *Statement - I:* The foot of the perpendicular from (3, 4) on the line $3x - 4y + 5 = 0$ is $(81/25, 92/25)$

Statement - II: If (h, k) is the foot of the perpendicular from (x_1, y_1) to the line

$$ax + by + c = 0 \text{ then } \frac{h - x_1}{a} = \frac{h - k_1}{b} = \frac{-(ax_1 + by_1 + c)}{a^2 + b^2}$$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
 2. Both Statements are true, Statement II is not correct explanation of Statement I.
 3. Statement I is true, Statement II is false.
 4. Statement I is false, Statement II is true.

LEVEL - 3**SINGLE CORRECT CHOICE TYPE:**

14. (0, 0) is the foot of the perpendicular from (4, 2) on a straight line. The equation of the line is
 1) $x - 2y = 0$ 2) $2x + y = 0$ 3) $x - y = 0$ 4) none of these
 15. If (2, -3) is the foot of the perpendicular from (-4, 5) on the line L, then the equation of the line L is
 1) $3x + 4y = 18$ 2) $4x - 3y = 18$ 3) $3x - 4y = 18$ 4) none of these
 16. The foot of the perpendicular from (1, 2) in a line L is $\left(\frac{-7}{5}, \frac{-6}{5}\right)$, then the equation of the line is
 1) $3x + 4y + 9 = 0$ 2) $3x - 4y = 1$ 3) $4x + 3y = 1$ 4) none of these

REASONING TYPE:

17. *Statement - I:* The reflection of (6, 8) in the line $x = y$ is (8, 6)
Statement - II: The reflection of (x_1, y_1) in the line $x = y$ is (y_1, x_1)
- Both Statements are true, Statement II is the correct explanation of Statement I.
 - Both Statements are true, Statement II is not correct explanation of Statement I.
 - Statement I is true, Statement II is false.
 - Statement I is false, Statement II is true.

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

18. P(2, 1) is foot of the point Q (4, 3) on the line
 1) $x + y = 3$ 2) $x - y = 1$ 3) $-x + 2y = 0$ 4) $3x - y = 5$
19. The image of (-4, 3) with respect to the line $7x - 24y + 50 = 0$ is
 1) $\left(\frac{72}{25}, \frac{-21}{25}\right)$ 2) $\left(\frac{-72}{25}, \frac{21}{25}\right)$ 3) $\left(\frac{-72}{25}, \frac{-21}{25}\right)$ 4) none of these
20. The line $x + y = 4$ is the perpendicular bisector of $\overline{AB}$ where A(3, -3), then B is
 1) (1, 7) 2) (7, 1) 3) (-1, 7) 4) (7, -1)

MATRIX MATCH TYPE:

- | 21. Column-I | Column-II |
|--|-------------------|
| a) Foot of the perpendicular from (3,4) to the line $3x - 4y = 18$ | 1) $(-7/5, -6/5)$ |
| b) Image of (-3,4) with respect to the origin | 2) $(-1, -14)$ |
| c) Image of (1,2) with respect to $3x + 4y - 1 = 0$ | 3) (6, 0) |
| d) The reflection of (4,-13) in the line $5x + y + 6 = 0$ | 4) (3, -4) |

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

22. The image of the point A(1, 2) by the line mirror $y = x$ is the point B and the image of B by line mirror $y = 0$ is the point
 1) (1, -2) 2) (0, 0) 3) (2, -1) 4) none of these
23. P (2, 1) is the image of the point Q (4, 3) about the line
 1) $x - y = 1$ 2) $2x - 3y = 0$ 3) $x + y = 0$ 4) none of these
24. The image of the point (4, 3) about the line $y = x$ is
 1) (3, 3) 2) (4, 4) 3) (3, 4) 4) none of these
25. The equation of the straight line equally inclined to the coordinate axes and equidistant from the point (1, -2) and (3, 4) is
 1) $x + y + 1 = 0$ 2) $x + y + 2 = 0$ 3) $x - y - 2 = 0$ 4) $x - y - 1 = 0$

Synopsis**Date: 30 & 31-08-2010****CONDITION FOR CONCURRENCY OF LINES :****Statement :**

If the lines $a_1x + b_1y + c = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent, then the condition is $a_3(b_1c_2 - b_2c_1) + b_3(c_1a_2 - c_2a_1) + c_3(a_1b_2 - a_2b_1) = 0$.

Proof :

$$\begin{aligned} \text{Given lines are} \quad a_1x + b_1y + c_1 &= 0 & \text{----- (1)} \\ a_2x + b_2y + c_2 &= 0 & \text{----- (2)} \\ a_3x + b_3y + c_3 &= 0 & \text{----- (3)} \end{aligned}$$

We know, the point of intersection of (1) and (2) is $\left[\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right]$

Since the three straight lines are concurrent, the point $\left[\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right]$ should lie on the third straight line.

$$\therefore \text{ The condition is } a_3 \left[\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1} \right] + b_3 \left[\frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right] + c_3 = 0$$

$$\Rightarrow a_3(b_1c_2 - c_1b_2) + b_3(c_1a_2 - c_2a_1) + c_3(a_1b_2 - a_2b_1) = 0.$$

$$\text{i.e., } \sum a_1(b_2c_3 - b_3c_2) = 0 \quad \text{or} \quad \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Statement :

If three constants p, q, r can be suitably found so that $p(a_1x + b_1y + c_1) + q(a_2x + b_2y + c_2) + r(a_3x + b_3y + c_3) = 0$ then the three given straight lines

$$\begin{aligned} a_1x + b_1y + c_1 &= 0 & \text{----- (1)} \\ a_2x + b_2y + c_2 &= 0 & \text{----- (2)} \\ a_3x + b_3y + c_3 &= 0 & \text{----- (3) are concurrent} \end{aligned}$$

Proof :

If P(h, k) is the point of intersection of the lines (1) and (2), we have $a_1h + b_1k + c_1 = 0$ and $a_2h + b_2k + c_2 = 0$.

Since given identity is true for all values of x and y, we have

$$p(a_1h + b_1k + c_1) + q(a_2h + b_2k + c_2) + r(a_3h + b_3k + c_3) = 0.$$

$$\Rightarrow r(a_3h + b_3k + c_3) = 0. \Rightarrow a_3h + b_3k + c_3 = 0. \text{ Since } r \neq 0.$$

$\therefore$ The point (h, k) lies on the third straight line.

Hence the three given straight lines are concurrent.

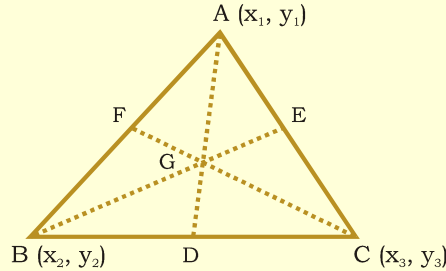
CENTROID:**Statement :**

The medians in a triangle are concurrent.

Proof :

Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of $\triangle ABC$. D, E, F be the midpoints of side $\overline{BC}$, $\overline{CA}$ and $\overline{AB}$ respectively.

$$\therefore D = \left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2} \right)$$



$$\therefore \text{Equation to the median } \overline{AD} \text{ is } y - y_1 = \frac{y_1 - \frac{y_2 + y_3}{2}}{x_1 - \frac{x_2 + x_3}{2}} (x - x_1)$$

$$\Rightarrow (y - y_1) (2x_1 - x_2 - x_3) = (2y_1 - y_2 - y_3) (x - x_1)$$

$$\Rightarrow L_1 \equiv (x - x_1) (2y_1 - y_2 - y_3) - (y - y_1) (2x_1 - x_2 - x_3) = 0 \quad \text{----- (1)}$$

Similarly equations to medians $\overline{BE}$ and $\overline{CF}$ are

$$L_2 \equiv (x - x_2) (2y_2 - y_3 - y_1) - (y - y_2) (2x_2 - x_3 - x_1) = 0 \quad \text{----- (2)}$$

$$\text{and } L_3 \equiv (x - x_3) (2y_3 - y_1 - y_2) - (y - y_3) (2x_3 - x_1 - x_2) = 0 \quad \text{----- (3)}$$

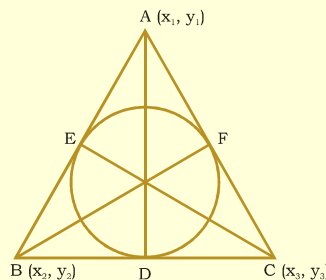
$$\text{Now } 1.L_1 + 1.L_2 + 1.L_3 = 0.$$

$\therefore$ The three medians $\overline{AD}$, $\overline{BE}$ and $\overline{CF}$ are concurrent.

INCENTRE:

Statement : Internal bisectors of the angles of a triangle are concurrent.

Proof : Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of a $\triangle ABC$.



Let $BC = a$, $CA = b$ and $AB = c$

Let the internal bisectors of the angles A, B, C meet $\overline{BC}$, $\overline{CA}$, $\overline{AB}$ in D, F, E respectively.

$$\therefore D \text{ divides } \overline{BC} \text{ in the ratio } AB : AC \Rightarrow BD : DC = c : b$$

$$\therefore D = \left(\frac{bx_2 + cx_3}{b+c}, \frac{by_2 + cy_3}{b+c} \right)$$

$$\text{Slope of } \overline{AD} = \frac{y_1 - \left(\frac{by_2 + cy_3}{b+c} \right)}{x_1 - \left(\frac{bx_2 + cx_3}{b+c} \right)} = \frac{(b+c)y_1 - (by_2 + cy_3)}{(b+c)x_1 - (bx_2 + cx_3)}$$

$$= \frac{b(y_1 - y_2) + c(y_1 - y_3)}{b(x_1 - x_2) + c(x_1 - x_3)}$$

$\therefore$ Equation to the angle bisector $\overline{AD}$

$$y - y_1 = \frac{b(y_1 - y_2) + c(y_1 - y_3)}{b(x_1 - x_2) + c(x_1 - x_3)} (x - x_1)$$

$$\Rightarrow (y - y_1) [b(x_1 - x_2) + c(x_1 - x_3)] = (x - x_1) [b(y_1 - y_2) + c(y_1 - y_3)]$$

$$L_1 \equiv (y - y_1) [b(x_1 - x_2) + c(x_1 - x_3)] - (x - x_1) [b(y_1 - y_2) + c(y_1 - y_3)] = 0$$

Similarly the equations of $\overline{BE}$ and $\overline{CF}$ are

$$L_2 \equiv (y - y_2) [c(x_2 - x_3) + a(x_2 - x_1)] - (x - x_2) [c(y_2 - y_3) + a(y_2 - y_1)] = 0$$

$$L_3 \equiv (y - y_3) [a(x_3 - x_1) + b(x_3 - x_2)] - (x - x_3) [a(y_3 - y_1) + b(y_3 - y_2)] = 0$$

We can easily verify that $aL_1 + bL_2 + cL_3 = 0 \Rightarrow L_1 = 0, L_2 = 0, L_3 = 0$ are concurrent.

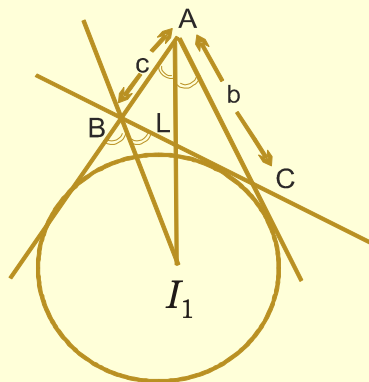
Hence the internal bisectors of the angles of a triangle are concurrent.

Excentre :

Co - ordinate of excentre opposite to $\angle A$ is given by

$$I_1 \equiv \left(\frac{-ax_1 + bx_2 + cx_3}{-a+b+c}, \frac{-ay_1 + by_2 + cy_3}{-a+b+c} \right) \quad (I_2 \text{ \& } I_3) \text{ opposite to AB and AC are given}$$

$$\text{by } I_2 \equiv \left(\frac{ax_1 - bx_2 + cx_3}{a-b+c}, \frac{ay_1 - by_2 + cy_3}{a-b+c} \right)$$



$$\frac{BL}{LC} = \frac{c}{b}, \text{ also } \frac{AI_1}{I_1L} = -\frac{b+c}{a}$$

$$I_3 \equiv \left(\frac{ax_1 + bx_2 - cx_3}{a+b-c}, \frac{ay_1 + by_2 - cy_3}{a+b-c} \right)$$

Note 1 :- The condition for the lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ to be concurrent is

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0$$

Note 2 :- The condition for the lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ are represent

$$\text{the same line, is } \frac{a_1}{a^2} = \frac{b_1}{b^2} = \frac{c_1}{c^2}$$

Illustration :

The incentre of the triangle whose sides have the equations

$x + y - 7 = 0$, $x - y + 1 = 0$ and $x - 3y + 5 = 0$ is

- 1) $(3, 1 + \sqrt{5})$ 2) $(4, 1 + \sqrt{5})$ 3) $(1, 2)$ 4) $(1 + \sqrt{5}, 3)$

Solution :

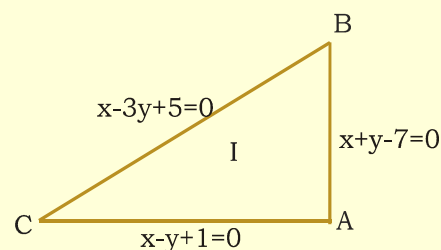
The given equations of the sides of a triangle are

$$x + y - 7 = 0 \quad \dots(1)$$

$$x - y + 1 = 0 \quad \dots(2)$$

$$x - 3y + 5 = 0 \quad \dots(3)$$

Solving these equation



we get the three co - ordinates of vertices of the triangle.

$$A \equiv (3, 4), B \equiv (4, 3) \text{ and } C \equiv (1, 2)$$

$$\text{also } AB = \sqrt{2}, BC = \sqrt{10} \text{ and } CA = 2\sqrt{2}$$

Hence the co - ordinate of in- center of the triangle is given by

$$\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c} \right) \equiv (3, 1 + \sqrt{5})$$

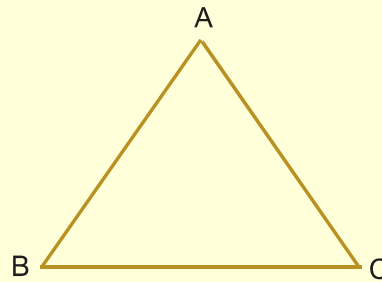
Illustrations :

The incentre I of $\triangle ABC$ if A is (4,-2), B is (-2,4) and C is (5,5) is

- 1) $\left(\frac{1}{2}, \frac{1}{2}\right)$ 2) $\left(\frac{3}{2}, \frac{3}{2}\right)$ 3) $\left(\frac{5}{2}, \frac{5}{2}\right)$ 4) $\left(\frac{7}{2}, \frac{7}{2}\right)$

Solution:

$$BC \equiv a = \sqrt{(5+2)^2 + (5-4)^2} = 5\sqrt{2}$$



$$CA \equiv b = \sqrt{(5-4)^2 + (5+4)^2} = 5\sqrt{2}$$

$$AB \equiv c = \sqrt{(-2-4)^2 + (4+2)^2} = 6\sqrt{2}$$

If incentre I is $(\bar{x}, \bar{y})$, then by the formula

$$\bar{x} = \frac{ax_1 + bx_2 + cx_3}{a + b + c} = \frac{20\sqrt{2} - 10\sqrt{2} + 30\sqrt{2}}{5\sqrt{2} + 5\sqrt{2} + 6\sqrt{2}} = \frac{5}{2}$$

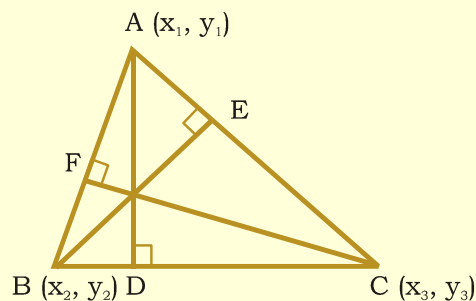
$$\bar{y} = \frac{ay_1 + by_2 + cy_3}{a + b + c} = \frac{-10\sqrt{2} + 20\sqrt{2} + 30\sqrt{2}}{5\sqrt{2} + 5\sqrt{2} + 6\sqrt{2}} = \frac{5}{2} \quad \therefore \text{ incentre is } \left(\frac{5}{2}, \frac{5}{2}\right)$$

ORTHOCENTRE:

Statement : The altitudes in a triangle are concurrent.

Proof : Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of $\triangle ABC$.

Let $\overline{AD}$, $\overline{BE}$ and $\overline{CF}$ be the altitudes.



$$\text{Slope of } \overline{BC} = \frac{y_2 - y_3}{x_2 - x_3}$$

$$\therefore \text{ Slope of } \overline{AD} = -\frac{x_2 - x_3}{y_2 - y_3}$$

$$\therefore \text{Equation to } \overline{AD} \text{ is } y - y_1 = -\frac{x_2 - x_3}{y_2 - y_3}(x - x_1)$$

$$\Rightarrow L_1 \equiv (x - x_1)(x_2 - x_3) + (y - y_1)(y_2 - y_3) = 0 \quad \text{----- (1)}$$

Similarly equations to $\overline{BE}$ and $\overline{CF}$ are

$$L_2 \equiv (x - x_2)(x_3 - x_1) + (y - y_2)(y_3 - y_1) = 0 \quad \text{----- (2)}$$

$$\text{and } L_3 \equiv (x - x_3)(x_1 - x_2) + (y - y_3)(y_1 - y_2) = 0 \quad \text{----- (3)}$$

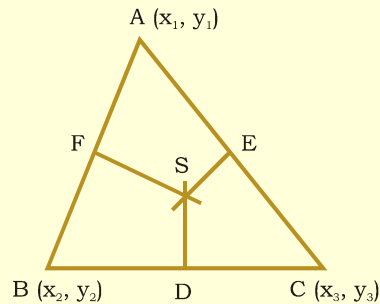
$$\text{Now } 1.L_1 + 1.L_2 + 1.L_3 = 0$$

$\therefore$ The three altitudes $\overline{AD}$, $\overline{BE}$ and $\overline{CF}$ are concurrent.

CIRCUMCENTRE:

Statement : The perpendicular bisectors of the sides of a triangle are concurrent.

Proof : Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ be the vertices of $\triangle ABC$.



If D, E, F are the midpoints of the sides $\overline{BC}$, $\overline{CA}$, $\overline{AB}$, then D =

$$\left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2} \right), E = \left(\frac{x_3 + x_1}{2}, \frac{y_3 + y_1}{2} \right), F = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right),$$

$$\text{Slope of } \overline{BC} = \frac{y_2 - y_3}{x_2 - x_3}$$

$$\therefore \text{Slope of perpendicular bisector} = -\frac{x_2 - x_3}{y_2 - y_3}$$

$\therefore$ Equation to the perpendicular bisector through D is

$$y - \frac{y_2 + y_3}{2} = -\frac{x_2 - x_3}{y_2 - y_3} \left(x - \frac{x_2 + x_3}{2} \right)$$

$$\Rightarrow 2y(y_2 - y_3) - (y_2^2 - y_3^2) = -2x(x_2 - x_3) + (x_2^2 - x_3^2)$$

$$L_1 \equiv 2x(x_2 - x_3) + 2y(y_2 - y_3) - (x_2^2 - x_3^2) - (y_2^2 - y_3^2) = 0$$

Similarly the perpendicular bisectors of $\overline{CA}$ and $\overline{AB}$ are

$$L_2 \equiv 2x(x_3 - x_1) + 2y(y_3 - y_1) - (x_3^2 - x_1^2) - (y_3^2 - y_1^2) = 0$$

$$L_3 \equiv 2x(x_1 - x_2) + 2y(y_1 - y_2) - (x_1^2 - x_2^2) - (y_1^2 - y_2^2) = 0$$

Now, $1.L_1 + 1.L_2 + 1.L_3 = 0 \Rightarrow L_1 = 0, L_2 = 0, L_3 = 0$ are concurrent.

Hence the perpendicular bisectors of the sides of $\triangle ABC$ are concurrent.

Illustrations :

The coordinates of the orthocentre of the triangle whose vertices are (0,0), (2,-1) and (-1,3) is

1) (-4,-3)

2) (-3,-4)

3) (1,2)

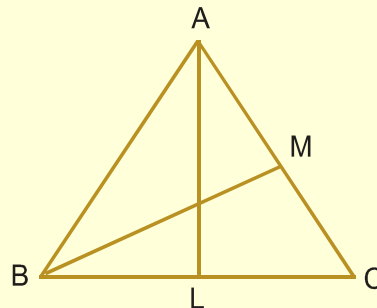
4) (4,3)

Solution:

Let A = (0,0), B = (2,-1) and C = (-1,3) $\left[\because \text{slope of } BC = \frac{-1-3}{2+1} \right]$

Since $AL \perp BC$, therefore equation of line AL is

$$y - 0 = \frac{3}{4}(x - 0)$$



or $4y = 3x$ or $3x - 4y = 0$

slope of AC = -3

Since $BM \perp AC$, therefore, equation of line BM is

$$y + 1 = \frac{1}{3}(x - 2) \text{ or } 3(y + 1) = x - 2$$

or $x - 3y = 5$

solving equations (i) and (ii),

we get $x = -4, y = -3$

$\therefore$ orthocentre (-4,-3)

Worksheet - 11, 12

Date: 30 & 31-08-2010

LEVEL - 1

SINGLE CORRECT CHOICE TYPE:

- The point of concurrency of medians of the triangle is _____
 1) Excentre 2) Centroid 3) Incentre 4) Orthocentre
- The incentre of the triangle formed by the lines $x = 0, y = 0, \frac{x}{3} + \frac{y}{4} = 1$ is
 1) (1, -1) 2) (1, 1) 3) (1, 2) 4) None

3. The orthocentre of the triangle formed by the lines $x - y = 0$, $x + y + 2 = 0$ and $3x - 4y + 12 = 0$ is
 1) (1, 1) 2) (1, 0) 3) (1, -1) 4) (-1, -1)
4. Two vertices of a triangle are (5, -1) and (-2, 3). If origin is the orthocentre, then the third vertex of the triangle is
 1) (4, -7) 2) (-4, 7) 3) (-4, -7) 4) (4, 7)
5. The circumcentre of the triangle with vertices (-2,3), (2,-1) and (4,0) is
 1) $\left(\frac{3}{2}, \frac{5}{2}\right)$ 2) $\left(\frac{3}{2}, -\frac{5}{2}\right)$ 3) $\left(-\frac{3}{2}, \frac{5}{2}\right)$ 4) $\left(-\frac{3}{2}, -\frac{5}{2}\right)$

MULTI CORRECT CHOICE TYPE:

6. If the lines $3x + y + 2 = 0$, $2x - y + 3 = 0$ and $a^2x + 2ay + 6 = 0$ are concurrent, then a is equal to
 1) $\sqrt{7} + 1$ 2) $\sqrt{7} - 1$ 3) $1 - \sqrt{7}$ 4) any real number

REASONING TYPE:

7. *Statement I* : The orthocentre of the triangle $\triangle ABC$ is (0,0) where $A = (0,0)$, $B = (a, 0)$, $C = (0,b)$
Statement II : In a right angle triangle the ortho centre is the vertex at right angle
- Both Statements are true, Statement II is the correct explanation of Statement I.
 - Both Statements are true, Statement II is not correct explanation of Statement I.
 - Statement I is true, Statement II is false.
 - Statement I is false, Statement II is true.

COMPREHENSION TYPE:

In an equilateral $\triangle^{le}$ orthocentre, circum centre and centroid coincide each other

8. If A(2,4) and B (2,6) are two vertices of an equilateral $\triangle^{le}$ then third vertex is
 1) $(\sqrt{3} - 2, 5)$ 2) $(5, 2 + \sqrt{3})$ 3) $(5, 2 - \sqrt{3})$ 4) $(2 + \sqrt{3}, 5)$
9. Orthocentre of equilateral $\triangle^{le} ABC$ is
 1) $\left(\frac{6 + \sqrt{3}}{3}, 5\right)$ 2) $\left(\frac{6 + \sqrt{3}}{3}, 6\right)$ 3) $\left(6 + \sqrt{3}, \frac{6 + \sqrt{3}}{3}\right)$ 4) $\left(\frac{6 - \sqrt{3}}{3}, \frac{5}{3}\right)$
10. Distance between orthocentre and circumcentre in the above triangle is
 1) 2 units 2) 1 unit 3) 3 units 4) 0 units

MATRIX MATCH TYPE:**11. Column-I****Column-II**

- a) Centroid of the triangle formed by $x + y - 7 = 0$, $x - y + 1 = 0$, $x - 3y + 5 = 0$ is 1) (5,5)
- b) Circumcentre of the triangle formed by $3x - y - 5 = 0$, $x + 2y - 4 = 0$, $5x + 3y + 1 = 0$ is 2) $(1/\sqrt{2}, 1/\sqrt{2})$
- c) Incentre of the triangle formed by $x + y = 1$, $x = 1$, $y = 1$ is 3) $(-6/7, 2/7)$
- d) Orthocentre of the triangle formed by $x - 2y + 9 = 0$, $x + y - 9 = 0$, $2x - y - 9 = 0$ is 4) $(8/3, 3)$

LEVEL - 2**SINGLE CORRECT CHOICE TYPE:**

12. If the lines $ax + by + c = 0$, $bx + cy + a = 0$ and $cx + ay + b = 0$ ($a \neq b \neq c$) are concurrent, then
- 1) $a^3 + b^3 + c^3 + 3abc = 0$ 2) $a^2 + b^2 + c^2 - 3abc = 0$
- 3) $a + b + c = 0$ 4) none of these

MULTI CORRECT CHOICE TYPE:

13. Three lines $px + qy + r = 0$, $qx + ry + p = 0$ and $rx + py + q = 0$ are concurrent if
- 1) $p + q + r = 0$ 2) $p^2 + q^2 + r^2 = pr + rp + pq$
- 3) $p^3 + q^3 + r^3 = 3pqr$ 4) None of these

LEVEL - 3**SINGLE CORRECT CHOICE TYPE:**

14. The lines $x - 3y - 6 = 0$, $3x + y - 4 = 0$, $\lambda x + 4y + \lambda = 0$ are concurrent if $\lambda =$
- 1) 1 2) 2 3) -1 4) -2
15. The mid points of the sides of a triangle are (5, 0), (5, 12) and (0, 12). The orthocenter of this triangle is

- 1) (0, 0) 2) (10, 0) 3) (0, 24) 4) $(\frac{13}{3}, 8)$

MULTI CORRECT CHOICE TYPE:

16. The sides of a triangle are $x + y = 1$, $7y = x$ and $\sqrt{3}y + x = 0$. Then the following is an interior point of the triangle .
- 1) circumcentre 2) centroid 3) incentre 4) orthocentre

LEVEL - 4**SINGLE CORRECT CHOICE TYPE:**

17. The straight line $x + 2y - 9 = 0$, $3x + 5y - 5 = 0$ and $ax + by - 1 = 0$ are concurrent if the straight line $35x - 22y + 1 = 0$ passes through the point
- 1) (a, b) 2) (b, a) 3) (a, -b) 4) (-a, b)
18. Find the circumcentre of the triangle formed by the points (2,1) (1, -2) and (-2, 3).

REASONING TYPE:

19. Statement - I : The orthocentre of the triangle formed by the lines

$$\frac{x}{a} + \frac{y}{b} = 1, y = 0 \text{ and } x = 0 \text{ lies on a vertex.}$$

Statement - II : Orthocentre is the point of intersection of altitudes.

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

LEVEL - 5**SINGLE CORRECT CHOICE TYPE:**

20. Find the circumcentre of the triangle formed by the lines $x + y = 0$, $2x + y = 5 = 0$ and $x - y = 2$.

COMPREHENSION TYPE:

The point of intersection of the bisectors of all the three angle bisectors of a triangle is called in-centre. The point of concurrency of the perpendicular bisectors of a triangle is called the circumcentre of the triangle. The point of concurrency of the three altitudes of a triangle is called its orthocentre.

Three sides of a triangle are ; $y = \sqrt{3}x$; $y = 0$ and $\sqrt{3}x + y = 2\sqrt{3}$

21. The incentre of the triangle is

- 1) (2,5) 2) (1,1) 3) (-1,-1) 4) $\left(1, \frac{1}{\sqrt{3}}\right)$

22. The circumcentre of the triangle is

- 1) $\left(1, \frac{\sqrt{3}}{3}\right)$ 2) (1,2) 3) (-2,-2) 4) none of these

23. The orthocentre of the triangle is

- 1) (0,0) 2) (2,0) 3) $(1, \sqrt{3})$ 4) $\left(1, \frac{1}{\sqrt{3}}\right)$